

**THE CONVEX NIVAT CONJECTURE:
A COMPLEXITY LOWER BOUND FOR STAR CONFIGURATIONS,
AND A REDUCTION THAT DEPENDS ON COLLE–GARIBALDI
AT ONE STEP ONLY**

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ABSTRACT. Nivat’s conjecture asserts that a configuration $\xi: \mathbb{Z}^2 \rightarrow \mathcal{A}$ over a finite alphabet with $P_\xi(m, n) \leq mn$ for some $m, n \geq 1$ is periodic. Its convex form replaces the rectangle by an arbitrary non-empty finite lattice-convex window S , that is, one with $S = \text{Conv}(S) \cap \mathbb{Z}^2$. This paper contains two results at two different levels, and it matters which is which.

The first is unconditional and self-contained, and occupies §§0–7: we call $\theta = \sum_{i=1}^m F_i: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ a *star configuration* when each F_i has a non-zero period $k_i v_i$, is not doubly periodic, and agrees with doubly periodic fields on the two half-planes $\{\pi_i < \ell_i\}$ and $\{\pi_i > r_i\}$, the primitive directions v_i being pairwise non-parallel; and we prove that $P_\theta(S) \geq |S| + 1$ for every non-empty finite lattice-convex S . The proof attaches to θ an *exceptional spectrum* Λ_i of roots of unity in each direction, a Laurent polynomial $A = \prod_i A_i(T^{v_i})$ that divides every annihilator of θ modulo constants, and the zonotope $Z = \text{Newt}(A) = \sum_i [0, d_i v_i]$. The zonotope then appears twice, with opposite signs: it is the exact cost of the affine relations carried by a window ($\dim U_S \geq |S| + 1 - |R_Z(S)|$), and it is the exact capacity for a family of quadratic characters that are independent modulo those relations. The two cancel, for every S and every Z .

The second result is a reduction (§8) of the convex conjecture to the first. Every link of the chain is either a published result (Kari–Szabados, Colle) or proved here in full, with a single exception: Proposition 3.1(ii) of the Colle–Garibaldi preprint [4] of June 2026, which has not been refereed, and which says that each periodic component is fully periodic on a half-plane on the far side as well. We carry it as Assumption 8.12, written into the hypothesis of Theorem 8.15. The other half of the structure theorem of [4] — full periodicity of each component on one half-plane — is proved here independently (Proposition 8.9), by a difference operator and a finite-state argument, with no appeal to minimality of the decomposition, to one-sided nonexpansive directions, or to modular minimality; and Lemma 8.14 turns any configuration of the resulting shape into a star configuration. Appendix D proves in full the $\mathbb{F}_p$ and convex form of Szabados’ theorem, quoted as [4, Theorem 2.8] and used in the proof of Proposition 3.1(ii) there, which the sources give only as a sketch. A further appendix reports a numerical verification of the main theorem and of the intermediate assertions of §§2–5, including a counterexample to the natural over-generalisation of Lemma 3.1.

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INTRODUCTION

The problem. Let $\mathcal{A}$ be a finite alphabet and $\xi: \mathbb{Z}^2 \rightarrow \mathcal{A}$ a configuration. For a finite $Q \subset \mathbb{Z}^2$ write $P_\xi(Q)$ for the number of distinct patterns $\xi|_{u+Q}$, $u \in \mathbb{Z}^2$, and $P_\xi(m, n) = P_\xi(\llbracket m \rrbracket \times \llbracket n \rrbracket)$ for the rectangular complexity, $\llbracket m \rrbracket = \{0, \dots, m-1\}$. The Morse–Hedlund theorem [11] says that a bi-infinite word is periodic as soon as its complexity function satisfies $P(n) \leq n$ for a single n . Nivat asked, in his 1997 ICALP address [12], whether the two-dimensional analogue holds:

if $P_\xi(m, n) \leq mn$ for some $m, n \geq 1$, is ξ periodic?

The question is open. Partial results fall into two families. On the combinatorial and dynamical side, Sander–Tijdeman [14], Epifanio–Koskas–Mignosi [7] and Quas–Zamboni [13] proved periodicity under complexity bounds of the form $P_\xi(m, n) \leq mn/C$; Cyr and Kra [5] brought the constant down to 2 by way of nonexpansive subdynamics and settled in [6] the short-rectangle case $P_\xi(k, n) \leq kn$ for $k \leq 3$; Colle and Garibaldi [3] gave a version corrected by the size of the alphabet. On the algebraic side, Kari and Szabados [8] observed that low complexity forces the existence of a non-trivial annihilating Laurent polynomial and deduced that the configuration decomposes as a sum of periodic configurations over $\mathbb{Z}$; Szabados [15] settled the case of two components; Colle [1, 2] developed the interaction between minimal periodic decompositions and one-sided nonexpansive directions, and Colle and Garibaldi [4] recently reached $k \leq 4$. Kari and Moutot [9] proved that the orbit closure of a configuration of low convex complexity contains a periodic configuration, and the related decidability statement for low-complexity tilings.

The *convex* form of the conjecture replaces the rectangle by an arbitrary window that is convex in the only sense available on a lattice: a finite $S \subset \mathbb{Z}^2$ is *lattice-convex* if $S = \text{Conv}(S) \cap \mathbb{Z}^2$. A configuration has *low convex pattern complexity* if $P_\xi(S) \leq |S|$ for some non-empty finite lattice-convex S . The convexity hypothesis is not decoration. Khetan [10] has constructed an exact cluster $F \subset \mathbb{Z}^2$ of cardinality 8 with full affine span and an F -tiling whose orbit closure contains no 1-periodic F -tiling; since every F -tiling has low complexity with respect to the window $-F$, this is a counterexample to the statement for non-convex windows of full affine span.

Results. Our results are two, and it matters which is which. The first is unconditional; the second is conditional on one citation, which is carried in its statement.

Call $\theta: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ a *star configuration* if $\theta = \sum_{i=1}^m F_i$ with $m \geq 2$, where each F_i is periodic in a direction v_i , is not doubly periodic, and coincides with doubly periodic fields on two complementary half-planes orthogonal to nothing in particular but bounded by the lines $\pi_i = \det(v_i, \cdot) = \text{const}$, the primitive directions $v_1, \dots, v_m$ being pairwise non-parallel. The precise axioms (S1)–(S3) are in §0.1. Informally: m one-dimensionally periodic fields, each of which is doubly periodic outside a strip, superimposed in m pairwise distinct directions.

Theorem A (Theorem 7.3). *Let θ be a star configuration and let $S \subset \mathbb{Z}^2$ be non-empty, finite and lattice-convex. Then $P_\theta(S) \geq |S| + 1$.*

Theorem A is proved in §§0–7 using no input from the literature beyond the facts that $\mathbb{C}[T_1^\pm, T_2^\pm]$ is a unique factorisation domain, that Newton polygons are additive under multiplication (Ostrowski), the Chinese remainder theorem, Hilbert’s Nullstellensatz, and the existence of unimodular triangulations of lattice polygons.

Theorem B (Theorem 8.15; conditional). *Assume Assumption 8.12 below, that is, the weak form of [4, Proposition 3.1(ii)]: for the configuration ϑ and the decomposition constructed in §8, every component is fully periodic on a half-plane V_i disjoint from the half-plane U_i already*

obtained. Let $\mathcal{A}$ be finite and $\xi: \mathbb{Z}^2 \rightarrow \mathcal{A}$. If $P_\xi(S) \leq |S|$ for some non-empty finite lattice-convex S , then ξ is periodic. Consequently, if $P_\xi(m, n) \leq mn$ for some $m, n \geq 1$, then ξ is periodic.

Theorem B is a reduction of the convex conjecture to Theorem A. The chain (§8) starts from the annihilator and the periodic decomposition of Kari and Szabados [8], passes through the two published results of Colle [1] (a line that is one-sidedly nonexpansive in both directions; a non-periodic configuration in the orbit closure which is fully periodic on a region), then through Proposition 8.9 of this paper (each component is fully periodic on one half-plane U_i), then through Assumption 8.12 (each component is fully periodic on a half-plane V_i on the far side), and finally through Lemma 8.14, which normalises such a configuration into a star configuration and contradicts Theorem A.

The one link that has not been refereed is Assumption 8.12. [4] is a preprint of June 2026 (v1); it proves that assertion in its Proposition 3.1(ii). We have gone through the relevant part of its dependency chain and record what we found (Remark 8.13 and the paragraphs following it), and Appendix D proves in full the one result used there which the literature gives only as a sketch, the $\mathbb{F}_p$ and convex form of Szabados' theorem; but we do *not* treat Assumption 8.12 as an established fact: it is written into the statement of Theorem B. The other half of the structure theorem of [4] — full periodicity of each component on one half-plane — has a complete independent proof here (Proposition 8.9), which uses neither minimality of the decomposition, nor one-sided nonexpansive directions, nor the modular minimality theorem of [2].

The shape of the proof of Theorem A. Let H_i be a period of F_i along v_i that also preserves every tail field, and let $D = \prod_{i=1}^m (T^{H_i} - 1)$.

A dichotomy (§1). Applying D to a function that depends on θ through a finite window kills everything except a bounded region, because outside the union of the pairwise intersections of m strips in pairwise distinct directions at most one component is still varying. If $D\mathcal{K}[\theta = a] \neq 0$ for some colour a , the theorem follows for *arbitrary* finite S , convex or not, by a one-line integral-domain argument. The rest of the paper is the remaining case $D\mathcal{K}[\theta = a] \equiv 0$.

The exceptional spectrum and its zonotope (§2). Pushing θ to infinity along $\pm H_i$ isolates the i -th component against a doubly periodic background, producing configurations $\Xi_i^\pm = F_i + B_i^\pm$ in the orbit closure together with their own pure tails $G_i^{\sigma, L}, G_i^{\sigma, R}$. The differences between them are invariant under H_i , hence decompose under the κ_i -th roots of unity; the set of roots that actually occur is the exceptional spectrum Λ_i , non-empty exactly because F_i is not doubly periodic. Set $A = \prod_i A_i(T^{v_i})$ with $A_i(t) = \prod_{\lambda \in \Lambda_i} (t - \lambda)$, and $Z = \text{Newt}(A) = \sum_i [0, d_i v_i]$, $d_i = |\Lambda_i|$. The key divisibility is Lemma 2.4: if $f(T)\eta$ is constant for a scalar encoding η of θ , then $A \mid f$. Counting, this says that a window S carries at most $|R_Z(S)|$ independent affine relations, where $R_Z(S) = \{r : r + Z \subseteq \text{Conv}(S)\}$: the *cost* of the zonotope.

The quadratic witness (§§3–6). Linear characters alone cannot close the gap, and they should not: the deficit $|R_Z(S)|$ is real. We produce $|R_Z(S)|$ further characters that are quadratic. For that one needs a single vector d with $D_z[\eta(z)\eta(z+d)] \neq 0$ and $d \in (Z - Z) \cap \mathbb{Z}^2$. Existence of some such d is §4; confinement to $Z - Z$ comes from a pair of recursions in the two variables (d, z) (§5) which say that the Laurent transform of the witness kills the ideal $(A(Y), A(X/Y))$ in $K[Y^\pm]$, $K = \mathbb{C}(X_1, X_2)$, whose quotient is spanned by monomials with exponents in $Z - Z$; and the passage from $d \in Z - Z$ to a *pair* of lattice points of Z differing by d is a unimodular triangulation argument (§6).

The cancellation (§7). Each $r \in R_Z(S)$ now yields a quadratic character Φ_r on the pattern set, and these are independent modulo the linear ones. The zonotope's capacity is exactly its cost, for every window and every zonotope; that is Remark 7.4.

Organisation. §0 fixes notation, the axioms of a star configuration, and the main theorem. §1 proves the dichotomy. §2 constructs Λ_i , A , Z and the dimension budget. §3 shows that A applied to the colour field of θ is a doubly periodic background. §4 produces a non-vanishing two-point difference, §5 confines it to $Z - Z$, and §6 converts that into a pair of lattice points of Z . §7 assembles Theorem A, and §8 deduces Theorem B: §8.1 sets up half-planes, regions and full periodicity; §8.2 collects the published input; §8.3 proves the first half-plane (Proposition 8.9); §8.4 states Assumption 8.12 and reports our reading of [4]; §8.5 normalises; §8.6 proves Theorem B. Appendix A is a one-page skeleton of the argument. Appendix B briefly describes, without proof and for the record only, a different route that was explored and is not used here. Appendix C reports the numerical verification. Appendix D proves the $\mathbb{F}_p$ and convex form of Szabados' theorem that [4, Theorem 2.8] uses.

Dependencies at a glance. §§0–7 are self-contained. The external input of §8 is Kari–Szabados [8] and Colle [1], both published; [2, Theorem 1.6] is used only to align the decomposition of this paper with the $\mathbb{Z}_p$ -minimal decomposition of [4]; and the only link that has not been refereed is Assumption 8.12, that is [4, Proposition 3.1(ii)]. Appendix D quotes nothing external beyond the one-dimensional Morse–Hedlund theorem.

0. NOTATION, HYPOTHESES AND THE MAIN THEOREM

0.1. Star configurations. Let p be a prime and $m \geq 2$. Let $v_1, \dots, v_m \in \mathbb{Z}^2$ be *primitive* (coprime coordinates) and *pairwise non-parallel*. Write

$$\pi_i(z) = \det(v_i, z) \quad (z \in \mathbb{Z}^2),$$

a surjective linear form $\mathbb{Z}^2 \rightarrow \mathbb{Z}$, surjective because v_i is primitive.

Call $G: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ *doubly periodic* if its period group $\text{Per}(G) = \{u \in \mathbb{Z}^2 : T^u G = G\}$ contains two linearly independent vectors; equivalently, $\text{Per}(G)$ is a finite-index sublattice of $\mathbb{Z}^2$.

Let $F_i: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ satisfy

- (S1) F_i has period $k_i v_i$ for some integer $k_i \geq 1$;
- (S2) F_i is *not* doubly periodic;
- (S3) there are doubly periodic fields L_i, R_i and integers $\ell_i \leq r_i + 1$ with

$$F_i = L_i \text{ on } \{\pi_i < \ell_i\}, \quad F_i = R_i \text{ on } \{\pi_i > r_i\}.$$

(If one starts with $\ell_i > r_i + 1$, shrinking ℓ_i to $r_i + 1$ keeps (S3).) We call $\{\pi_i < \ell_i\}$ and $\{\pi_i > r_i\}$ the *left* and *right tail regions* of component i , call L_i, R_i its *left* and *right tails*, and call $\{\ell_i \leq \pi_i \leq r_i\}$ the *transition strip* (possibly empty). We do not require $L_i = R_i$, do not require either to be constant, and do not require $k_i = 1$. Put

$$\theta = \sum_{i=1}^m F_i: \mathbb{Z}^2 \rightarrow \mathbb{F}_p,$$

and call θ a *star configuration*.

0.2. Complexity and the main theorem. For finite $Q \subset \mathbb{Z}^2$ let $\theta|_{u+Q}$ denote the function $q \mapsto \theta(u+q)$ and

$$P_\theta(Q) = \#\{\theta|_{u+Q} : u \in \mathbb{Z}^2\}.$$

Call a finite $S \subset \mathbb{Z}^2$ *lattice-convex* if $S = \text{Conv}(S) \cap \mathbb{Z}^2$.

Theorem T. *Let θ be a star configuration and let $S \subset \mathbb{Z}^2$ be non-empty, finite and lattice-convex. Then*

$$P_\theta(S) \geq |S| + 1.$$

Remark 0.1 (Periodic backgrounds). If a doubly periodic field B is added to θ , absorb B into F_1 and replace the tangential period of F_1 by a multiple that also preserves B ; such a multiple exists because $\text{Per}(B)$ has finite index, so $\text{Per}(B) \cap \mathbb{Z}k_1v_1 \neq 0$. Then $F_1 + B$ is still not doubly periodic (otherwise $F_1 = (F_1 + B) - B$ would be), and $L_1 + B$, $R_1 + B$ are still doubly periodic. Hence Theorem T covers an arbitrary doubly periodic background.

0.3. Translations, differences and basic conventions. Write $T^u f(z) = f(z + u)$. A *Laurent polynomial* is a finite sum $f = \sum_u c_u T^u$ with complex coefficients unless stated otherwise, with support $\text{supp}(f) = \{u : c_u \neq 0\}$ and Newton polygon $\text{Newt}(f) = \text{Conv}(\text{supp}(f))$. The ring $\mathbb{C}[T_1^\pm, T_2^\pm]$ is an integral domain and a unique factorisation domain, and its units are exactly the monomials cT^u with $c \in \mathbb{C}^\times$.

Colour indicators, the encoding, the differences and all linear algebra take place in characteristic zero; only the summation of the original components happens in $\mathbb{F}_p$. In particular $\mathcal{K}[\theta = a]$ takes values in $\{0, 1\} \subset \mathbb{Z}$.

The common period lattice and the vectors H_i . Put

$$\Gamma := \bigcap_{j=1}^m (\text{Per}(L_j) \cap \text{Per}(R_j)),$$

a finite intersection of finite-index sublattices, hence of finite index; choose $N \in \mathbb{Z}_{>0}$ with $N\mathbb{Z}^2 \subseteq \Gamma$. For each i we have $Nk_i v_i \in \Gamma$, so we may put

$$H_i := \kappa_i v_i, \quad \kappa_i := \min\{\kappa \in k_i \mathbb{Z}_{>0} : \kappa v_i \in \Gamma\}.$$

Thus H_i preserves F_i (by (S1)) and preserves every L_j, R_j , $j = 1, \dots, m$.

The difference operator and its formal exponents. Put

$$D = \prod_{i=1}^m (T^{H_i} - 1), \quad I_a = \mathcal{K}[\theta = a], \quad e_\theta = (I_a)_{a \in \mathbb{F}_p}.$$

Then $D \neq 0$: each factor is non-zero and $\mathbb{C}[T^\pm]$ is an integral domain. For $C \subseteq [m] := \{1, \dots, m\}$ write $H_C = \sum_{i \in C} H_i$, and call

$$\mathcal{E} = \{H_C : C \subseteq [m]\}, \quad \mathcal{E}_i = \{H_C : C \subseteq [m] \setminus \{i\}\}$$

the *formal exponent sets*. For an arbitrary function g one has the formal expansion

$$(Dg)(z) = \sum_{C \subseteq [m]} (-1)^{m-|C|} g(z + H_C), \quad (0.1)$$

valid as an operator identity irrespective of whether the vectors H_C for different C coincide. Thus $\text{supp}(D) \subseteq \mathcal{E}$, and for $m \geq 3$ the inclusion may be strict, since distinct subsets may have equal sums (for instance if $H_3 = H_1 + H_2$). *Every pairing argument made on a formal expansion below is indexed by subsets C , never by the points H_C .* Likewise, for $Q_i := \prod_{j \neq i} (T^{H_j} - 1)$ one has $D = (T^{H_i} - 1)Q_i$ and

$$(Q_i g)(z) = \sum_{C \subseteq [m] \setminus \{i\}} (-1)^{m-1-|C|} g(z + H_C). \quad (0.2)$$

1. FINITE DIFFERENCES AND THE FIRST CASE

1.1. Differencing turns a local function into one of finite support.

Lemma 1.1. *Let $h(z) = \varphi(\theta|_{z+W})$ with $W \subset \mathbb{Z}^2$ finite. Then Dh has finite support.*

Proof. Put $M = \mathcal{E} + W$, a finite set, and $\text{diam}_i = \max_{s \in M} |\pi_i(s)|$. Define the *widened strips*

$$\Sigma_i = \{z : \ell_i - \text{diam}_i \leq \pi_i(z) \leq r_i + \text{diam}_i\},$$

and call direction i *active at the anchor* z if $z \in \Sigma_i$. If i is not active at z , then for every $s \in M$ the values $\pi_i(z+s)$ and $\pi_i(z)$ lie both in $\{\pi_i < \ell_i\}$ or both in $\{\pi_i > r_i\}$, so on $z+M$ the field F_i is identically one pure tail $\varepsilon_i \in \{L_i, R_i\}$.

Claim: if at most one direction is active at z , then $Dh(z) = 0$. Let J be the set of active directions, $|J| \leq 1$. Take $j_0 \in J$ if $J \neq \emptyset$, and $j_0 = 1$ otherwise. On $z+M$,

$$\theta = G + B, \quad G = \begin{cases} F_{j_0}, & J = \{j_0\} \\ 0, & J = \emptyset \end{cases}, \quad B = \sum_{j \notin J} \varepsilon_j,$$

where B is doubly periodic and preserved by every H_l , and G is preserved by H_{j_0} by (S1). Hence $G+B$ is preserved by H_{j_0} . Expand $D = (T^{H_{j_0}} - 1)Q_{j_0}$ using (0.2):

$$Dh(z) = \sum_{C \subseteq [m] \setminus \{j_0\}} (-1)^{m-1-|C|} [h(z + H_C + H_{j_0}) - h(z + H_C)].$$

Fix C . Since H_C and $H_C + H_{j_0}$ both lie in $\mathcal{E}$, the two windows $z + H_C + W$ and $z + H_C + H_{j_0} + W$ are contained in $z + M$, where $\theta = G + B$. So for $y \in W$,

$\theta(z + H_C + H_{j_0} + y) = (G + B)(z + H_C + y + H_{j_0}) = (G + B)(z + H_C + y) = \theta(z + H_C + y)$, so the two W -patterns agree and the bracket vanishes. This proves the claim.

Conclusion. Therefore $\text{supp}(Dh) \subseteq \bigcup_{i \neq j} (\Sigma_i \cap \Sigma_j)$. As $v_i \nparallel v_j$, the intersection of two strips of bounded width in non-parallel directions is bounded, so Dh has finite support. $\square$

1.2. The integral-domain argument.

Lemma 1.2. *Let $J: \mathbb{Z}^2 \rightarrow \mathbb{C}$ be non-zero with finite support and let f be a non-zero Laurent polynomial. Then $f(T)J \neq 0$.*

Proof. Write $\hat{J}(X) = \sum_z J(z)X^{-z} \in \mathbb{C}[X_1^\pm, X_2^\pm]$, which is non-zero. For $f = \sum_u c_u T^u$,

$$\sum_z (f(T)J)(z)X^{-z} = \sum_u c_u \sum_z J(z+u)X^{-z} = \sum_u c_u \sum_{z'} J(z')X^{-z'+u} = \left(\sum_u c_u X^u\right) \hat{J}(X).$$

Both factors are non-zero Laurent polynomials and the ring is an integral domain, so the product is non-zero. $\square$

1.3. The first case.

Proposition 1.3. *If $DI_a \neq 0$ for some $a \in \mathbb{F}_p$, then $P_\theta(S) \geq |S| + 1$ for every non-empty finite $S \subset \mathbb{Z}^2$; no convexity is needed.*

Proof. Suppose $P_\theta(S) \leq |S|$. Write $\mathcal{L} = \{\theta|_{u+S} : u \in \mathbb{Z}^2\}$, so $|\mathcal{L}| \leq |S|$. On $\mathcal{L}$ consider the $|S| + 1$ functions

$$\mathbf{1}; \quad \gamma \mapsto \mathbb{K}[\gamma(s) = a] \quad (s \in S).$$

They lie in a space of dimension $\leq |\mathcal{L}| \leq |S|$, hence are linearly dependent: there are rationals $c_0, (c_s)_{s \in S}$, not all zero, with

$$\sum_{s \in S} c_s \mathbb{K}[\gamma(s) = a] = c_0 \quad \text{for all } \gamma \in \mathcal{L}.$$

Substituting $\gamma = \theta|_{u+S}$ gives $q(T)I_a = c_0$ with $q = \sum_s c_s T^s$. If $q = 0$ then all $c_s = 0$ and hence $c_0 = 0$, a contradiction; so $q \neq 0$.

Apply D to both sides. The right-hand side gives $Dc_0 = 0$, since a constant is killed by $T^{H_1} - 1$; the left-hand side gives $q(T)DI_a$. By Lemma 1.1 and the hypothesis, DI_a is non-zero with finite support, so $q(T)DI_a \neq 0$ by Lemma 1.2. Contradiction. $\square$

From now on we assume

$$DI_a = 0 \quad \text{for every } a \in \mathbb{F}_p. \quad (1.1)$$

Consequently $Dw(\theta) = \sum_a w(a)DI_a = 0$ for every function $w: \mathbb{F}_p \rightarrow \mathbb{Z}$.

Remark 1.4 (The dichotomy does not depend on the choice of the H_i). Replacing H_i by a positive multiple $n_i H_i$ produces $D' = \prod_i (T^{n_i H_i} - 1)$, which is a multiple of D because $T^{nH} - 1 = (T^H - 1)(1 + T^H + \dots + T^{(n-1)H})$. Hence $DI_a = 0$ implies $D'I_a = 0$; conversely, if $DI_a \neq 0$ then it is non-zero of finite support by Lemma 1.1, and Lemma 1.2 gives $D'I_a = (D'/D)(T)DI_a \neq 0$. So whether (1.1) holds does not depend on the particular κ_i chosen, as long as H_i preserves all the tail fields.

2. EXCEPTIONAL SPECTRA, THE ENCODING, AND THE DIMENSION BUDGET FOR AFFINE RELATIONS

2.1. Isolating configurations. Fix i . For $j \neq i$ we have $\pi_j(H_i) = \det(v_j, H_i) = \kappa_i \det(v_j, v_i) \neq 0$, so $\pi_j(nH_i) \rightarrow \pm\infty$; hence on any fixed finite set F_j is eventually identically equal, as $n \rightarrow +\infty$, to

$$\varepsilon_j^+ = \begin{cases} R_j, & \pi_j(H_i) > 0, \\ L_j, & \pi_j(H_i) < 0, \end{cases}$$

while F_i is preserved by H_i . So $T^{nH_i}\theta$ stabilises on every finite set, with limit

$$\Xi_i^+ = F_i + B_i^+, \quad B_i^+ = \sum_{j \neq i} \varepsilon_j^+.$$

Letting $n \rightarrow -\infty$ gives $\Xi_i^- = F_i + B_i^-$, with the opposite tail choices: ε_j^- is L_j when $\pi_j(H_i) > 0$ and R_j otherwise. Both belong to the orbit closure $X_\theta = \overline{\{T^u\theta : u \in \mathbb{Z}^2\}}$ in the product topology. Since $\kappa_i > 0$ we have $\text{sign } \pi_j(H_i) = \text{sign } \pi_j(v_i)$, so that

$$\varepsilon_j^\sigma = \begin{cases} R_j, & \sigma\pi_j(v_i) > 0, \\ L_j, & \sigma\pi_j(v_i) < 0. \end{cases} \quad (2.0)$$

By (S3),

$$\Xi_i^\sigma = G_i^{\sigma,L} := L_i + B_i^\sigma \quad \text{on } \{\pi_i < \ell_i\}, \quad \Xi_i^\sigma = G_i^{\sigma,R} := R_i + B_i^\sigma \quad \text{on } \{\pi_i > r_i\},$$

and $G_i^{\sigma,L}, G_i^{\sigma,R}$ are doubly periodic and preserved by Γ . They too belong to X_θ : take (v_i, u_i) a basis of $\mathbb{Z}^2$ with $\det(v_i, u_i) = 1$ and put $u = Nu_i \in \Gamma$, so that $\pi_i(u) = N > 0$. For fixed z and n large, $\pi_i(z + nu) > r_i$, whence

$$(T^{nu}\Xi_i^\sigma)(z) = R_i(z + nu) + B_i^\sigma(z + nu) = R_i(z) + B_i^\sigma(z) = G_i^{\sigma,R}(z),$$

that is, $T^{nu}\Xi_i^\sigma \rightarrow G_i^{\sigma,R}$; using $-u$ instead gives $G_i^{\sigma,L}$.

We record that H_i preserves Ξ_i^σ , and every H_j preserves each of $G_i^{\sigma,L}, G_i^{\sigma,R}$, the latter being sums of tails.

2.2. Spectral projections, exceptional difference fields, and Λ_i . Take an integral basis (v_i, u_i) with $\det(v_i, u_i) = 1$ and write $z = sv_i + tu_i$; then $\pi_i(z) = t$ and $s = \det(z, u_i)$. Call $\{t = \text{const}\}$ a *transverse row*. Recall $H_i = \kappa_i v_i$.

Lemma 2.0 (Spectral projection). *Let $d: \mathbb{Z}^2 \rightarrow \mathbb{C}$ satisfy $T^{H_i}d = d$. For $\lambda^{\kappa_i} = 1$ put*

$$P_\lambda := \frac{1}{\kappa_i} \sum_{k=0}^{\kappa_i-1} \lambda^{-k} T^{kv_i}, \quad \widehat{d}_t(\lambda) := \frac{1}{\kappa_i} \sum_{k=0}^{\kappa_i-1} \lambda^{-k} d(kv_i + tu_i).$$

Then:

- (1) $(P_\lambda d)(sv_i + tu_i) = \lambda^s \widehat{d}_t(\lambda)$, and $\sum_\lambda P_\lambda d = d$;
- (2) $T^{v_i} P_\lambda d = \lambda P_\lambda d$;
- (3) P_λ is a Laurent polynomial in T , hence commutes with every T^u and with every Laurent polynomial $f(T)$;
- (4) for $f(T) = \sum_{\alpha, \beta} f_{\alpha\beta} T^{\alpha v_i + \beta u_i}$,

$$(f(T)P_\lambda d)(sv_i + tu_i) = \lambda^s \sum_{\alpha, \beta} f_{\alpha\beta} \lambda^\alpha \widehat{d}_{t+\beta}(\lambda) = \lambda^s [f(\lambda, Y) \widehat{d}_\bullet(\lambda)](t),$$

where $f(\lambda, Y) := \sum_{\alpha, \beta} f_{\alpha\beta} \lambda^\alpha Y^\beta$ acts, as a Laurent polynomial in the single variable Y , on functions of t , with Y the shift $t \mapsto t + 1$.

Proof. Parts 1 and 2 are the discrete Fourier decomposition on $\mathbb{Z}/\kappa_i$, carried out row by row: $(P_\lambda d)(sv_i + tu_i) = \frac{1}{\kappa_i} \sum_k \lambda^{-k} d((s+k)v_i + tu_i) = \lambda^s \cdot \frac{1}{\kappa_i} \sum_{k'} \lambda^{-k'} d(k'v_i + tu_i)$, re-indexing by κ_i -periodicity, while $\sum_\lambda \lambda^{s-k} = \kappa_i \mathbb{1}[k \equiv s]$ gives completeness. Part 3 is clear, and part 4 is a direct computation from part 1 and the definition of $T^{\alpha v_i + \beta u_i}$. $\square$

Say that λ occurs in d if $P_\lambda d \neq 0$, that is, if $\widehat{d}_t(\lambda) \neq 0$ for some row t .

Define the *exceptional difference fields*

$$d_{i, \sigma, \epsilon, a}(z) = \mathbb{1}[\Xi_i^\sigma(z) = a] - \mathbb{1}[G_i^{\sigma, \epsilon}(z) = a] \in \{-1, 0, 1\} \quad (\sigma \in \{+, -\}, \epsilon \in \{L, R\}, a \in \mathbb{F}_p).$$

(P1) $d_{i, \sigma, R, a} \equiv 0$ on $\{\pi_i > r_i\}$, and $d_{i, \sigma, L, a} \equiv 0$ on $\{\pi_i < \ell_i\}$.

(P2) Both Ξ_i^σ and $G_i^{\sigma, \epsilon}$ are preserved by H_i , so $T^{H_i} d_{i, \sigma, \epsilon, a} = d_{i, \sigma, \epsilon, a}$ and Lemma 2.0 applies.

Put

$$\Lambda_i = \{\lambda : \lambda^{\kappa_i} = 1, \exists \sigma, \epsilon, a \text{ such that } \lambda \text{ occurs in } d_{i, \sigma, \epsilon, a}\}.$$

Lemma 2.1. $\Lambda_i \neq \emptyset$.

Proof. Otherwise every $d_{i, \sigma, \epsilon, a}$ vanishes identically, by part 1 of Lemma 2.0. Taking $\sigma = +$ and $\epsilon = R$ gives $\Xi_i^+ = G_i^{+, R}$; subtracting B_i^+ gives $F_i = R_i$, so F_i is doubly periodic, contradicting (S2). $\square$

Put

$$A_i(t) = \prod_{\lambda \in \Lambda_i} (t - \lambda), \quad d_i = \deg A_i = |\Lambda_i| \geq 1, \quad (2.1)$$

$$A(T) = \prod_{i=1}^m A_i(T^{v_i}), \quad Z = \sum_{i=1}^m [0, d_i v_i] \subset \mathbb{R}^2. \quad (2.2)$$

(P3) For all σ, ϵ, a one has $A_i(T^{v_i}) d_{i, \sigma, \epsilon, a} = 0$. Indeed, by Lemma 2.0 only the terms with $\lambda \in \Lambda_i$ in $d = \sum_\lambda P_\lambda d$ are non-zero, and $A_i(T^{v_i}) P_\lambda d = A_i(\lambda) P_\lambda d = 0$.

The elements of Λ_i are distinct κ_i -th roots of unity, so A_i is monic with $A_i(0) = \prod_{\lambda} (-\lambda) \neq 0$. Hence

$$\text{Newt}(A_i(T^{v_i})) = [0, d_i v_i], \quad \text{Newt}(A) = \sum_i [0, d_i v_i] = Z, \quad (2.3)$$

the second equality by Minkowski additivity of Newton polygons (Ostrowski: over an integral domain, $\text{Newt}(fg) = \text{Newt}(f) + \text{Newt}(g)$).

Remark 2.1' (Galois stability of Λ_i). Each $d_{i,\sigma,\epsilon,a}$ is integer-valued, so for every $\tau \in \text{Gal}(\mathbb{Q}(\zeta_{\kappa_i})/\mathbb{Q})$ one has $\tau(\widehat{d}_t(\lambda)) = \widehat{d}_t(\tau\lambda)$. Hence λ occurs if and only if $\tau\lambda$ occurs, Λ_i is a union of Galois orbits, A_i is a product of distinct cyclotomic polynomials, $A_i \in \mathbb{Z}[t]$ and $A \in \mathbb{Z}[T^\pm]$. Nothing below depends on this (Lemma 2.5 handles the coefficient field by invariance of rank under field extension), but it does mean that every object of §2 can be discussed over $\mathbb{Q}$.

2.3. A spectrum-preserving encoding.

Lemma 2.2. *There is an injection $w: \mathbb{F}_p \rightarrow \mathbb{Z}$ such that for every i and every $\lambda \in \Lambda_i$, the scalar difference field*

$$\delta_{i,\sigma,\epsilon} := w(\Xi_i^\sigma) - w(G_i^{\sigma,\epsilon}) = \sum_a w(a) d_{i,\sigma,\epsilon,a}$$

still has λ occurring in it, for some pair σ, ϵ .

Proof. Regard $(w(a))_{a \in \mathbb{F}_p}$ as a vector in $\mathbb{Q}^p$. For each $\lambda \in \Lambda_i$ there are, by definition, σ, ϵ, t for which the vector $(\widehat{d_{i,\sigma,\epsilon,a}})_t(\lambda) \in \mathbb{C}^p$, indexed by a , is non-zero. By linearity of the Fourier transform,

$$(\widehat{\delta_{i,\sigma,\epsilon}})_t(\lambda) = \sum_a w(a) (\widehat{d_{i,\sigma,\epsilon,a}})_t(\lambda)$$

is a non-zero complex linear form in w ; its zero set inside $\mathbb{Q}^p$ is a proper $\mathbb{Q}$ -subspace of $\mathbb{Q}^p$, since otherwise the form would vanish on a $\mathbb{Q}$ -basis of $\mathbb{Q}^p$ and hence identically. There are $\sum_i |\Lambda_i|$ such conditions. Injectivity excludes a further $\binom{p}{2}$ proper hyperplanes $\{w(a) = w(a')\}$. Finitely many proper subspaces cannot cover $\mathbb{Q}^p$, so a rational w satisfying all the conditions exists; multiplying by a common denominator makes it integer-valued without affecting non-vanishing or injectivity. $\square$

Fix such a w and put $\eta = w(\theta): \mathbb{Z}^2 \rightarrow \mathbb{Z}$. By (1.1), $D\eta = 0$.

2.4. One-sided vanishing.

Lemma 2.3. *Let $d: \mathbb{Z}^2 \rightarrow \mathbb{C}$ satisfy $T^{H_i}d = d$, vanish identically on a half-plane $\{\pi_i > c\}$ or on a half-plane $\{\pi_i < c\}$, and let λ occur in d . If a Laurent polynomial f satisfies $f(T)d = 0$, then $(T^{v_i} - \lambda) \mid f$.*

Proof. By part 3 of Lemma 2.0, $f(T)P_\lambda d = P_\lambda f(T)d = 0$. Write $c_\lambda(t) = \widehat{d}_t(\lambda)$, so $c_\lambda(t) = 0$ whenever d vanishes on row t , and $g(Y) := f(\lambda, Y) = \sum_\beta g_\beta Y^\beta$. By part 4 of Lemma 2.0,

$$0 = (f(T)P_\lambda d)(sv_i + tu_i) = \lambda^s \sum_\beta g_\beta c_\lambda(t + \beta) \quad \text{for all } s, t,$$

that is, $g(Y)c_\lambda = 0$; and $c_\lambda \not\equiv 0$ because λ occurs.

Case: d vanishes on $\{\pi_i > c\}$. Then $c_\lambda(t) = 0$ for $t > c$, so $t_0 := \max\{t : c_\lambda(t) \neq 0\}$ exists. If $g \neq 0$, let $\beta_- = \min\{\beta : g_\beta \neq 0\}$ and evaluate at $t = t_0 - \beta_-$: for $\beta > \beta_-$ one has $t_0 - \beta_- + \beta > t_0$ and c_λ vanishes there, so only the term $\beta = \beta_-$ survives, giving $g_{\beta_-} c_\lambda(t_0) = 0$, a contradiction.

Case: d vanishes on $\{\pi_i < c\}$. Symmetrically, take $t_0 = \min\{t : c_\lambda(t) \neq 0\}$ and $\beta_+ = \max\{\beta : g_\beta \neq 0\}$ and evaluate at $t = t_0 - \beta_+$; only the term $\beta = \beta_+$ survives.

So $g \equiv 0$. Write f as an element of $\mathbb{C}[X^\pm, Y^\pm]$ with $X = T^{v_i}$ and $Y = T^{u_i}$; since (v_i, u_i) is a basis of $\mathbb{Z}^2$ this is an automorphism of the Laurent ring. With $f = \sum_\beta Y^\beta f_\beta(X)$, the identity $g \equiv 0$ says $f_\beta(\lambda) = 0$ for every β , whence $(X - \lambda) \mid f_\beta(X)$ — a one-variable Laurent polynomial vanishing at a non-zero point is divisible by $X - \lambda$ — and therefore $(X - \lambda) \mid f$. $\square$

2.5. Divisibility.

Lemma 2.4. *Let f be a non-zero Laurent polynomial with complex coefficients such that $f(T)\eta = c$ is constant. Then $A \mid f$.*

Proof. (a) *The relation passes to the orbit closure.* The map $z \mapsto (f(T)\eta)(z) - c$ depends on θ only through its values on $z + \text{supp}(f)$, so it is a local function of θ . It vanishes on every translate of θ ; local functions are continuous for the product topology, so it vanishes on every member χ of X_θ : $f(T)w(\chi) = c$.

(b) *Subtraction.* Take $\chi = \Xi_i^\sigma$ and $\chi = G_i^{\sigma, \epsilon}$, both in X_θ by §2.1, and subtract the two identities: $f(T)\delta_{i, \sigma, \epsilon} = 0$.

(c) *One prime factor at a time.* Fix $\lambda \in \Lambda_i$. By Lemma 2.2 there are σ, ϵ with λ occurring in $\delta_{i, \sigma, \epsilon}$; by (P1) this difference field vanishes on a half-plane; by (P2) it is preserved by H_i . Lemma 2.3 gives $T^{v_i} - \lambda \mid f$.

(d) *Assembling.* Under the automorphism $X = T^{v_i}$, $Y = T^{u_i}$ the element $T^{v_i} - \lambda$ becomes $X - \lambda$, and $\mathbb{C}[X^\pm, Y^\pm]/(X - \lambda) \cong \mathbb{C}[Y^\pm]$ is an integral domain, so $T^{v_i} - \lambda$ is prime. Primes coming from different directions $i \neq k$ are non-associate: being associate would force $T^{v_i} - \lambda = cT^u(T^{v_k} - \mu)$, and comparing supports gives $\{0, v_i\} = \{u, u + v_k\}$, hence $v_i = \pm v_k$, contradicting non-parallelism. Primes from the same direction with $\lambda \neq \lambda'$ are also non-associate: comparing supports forces $u = 0$ and then comparing coefficients forces $\lambda = \lambda'$. In a unique factorisation domain, pairwise non-associate primes each dividing f have their product dividing f : $A = \prod_i \prod_{\lambda \in \Lambda_i} (T^{v_i} - \lambda) \mid f$. $\square$

2.6. The dimension budget for affine relations. For a lattice-convex S put

$$R_Z(S) = \{r \in \mathbb{Z}^2 : r + Z \subseteq \text{Conv}(S)\},$$

$$U_S = \text{span}_{\mathbb{Q}} \left\{ \mathbf{1}, \quad \gamma \mapsto w(\gamma(s)) \ (s \in S) \right\} \subseteq \{\mathcal{L} \rightarrow \mathbb{Q}\}.$$

Lemma 2.5. $\dim_{\mathbb{Q}} U_S \geq |S| - |R_Z(S)| + 1$.

Proof. U_S is generated by $|S| + 1$ elements, so $\dim_{\mathbb{Q}} U_S = (|S| + 1) - \dim_{\mathbb{Q}} \mathcal{R}$, where

$$\mathcal{R} = \left\{ (c_0, (c_s)_{s \in S}) \in \mathbb{Q}^{1+|S|} : \sum_{s \in S} c_s \eta(u + s) = c_0 \text{ for all } u \in \mathbb{Z}^2 \right\}.$$

$\mathcal{R}$ is the $\mathbb{Q}$ -solution space of a (possibly infinite) homogeneous linear system with integer coefficients; writing $\mathcal{R}_{\mathbb{C}}$ for its $\mathbb{C}$ -solution space, $\mathcal{R}_{\mathbb{C}} = \mathcal{R} \otimes_{\mathbb{Q}} \mathbb{C}$ and $\dim_{\mathbb{C}} \mathcal{R}_{\mathbb{C}} = \dim_{\mathbb{Q}} \mathcal{R}$, the dimension of the solution space of a rational linear system being unchanged by extension of the base field.

Let $(c_0, (c_s)) \in \mathcal{R}_{\mathbb{C}}$ be non-zero and put $f = \sum_s c_s T^s$. If $f = 0$ then $c_0 = 0$ and the element is zero; so $f \neq 0$, and $f(T)\eta = c_0$ with $\text{supp}(f) \subseteq S$. By Lemma 2.4, valid for complex f , we may write $f = Ag$ with $g \neq 0$. By (2.3) and Newton-polygon additivity,

$$\text{Newt}(f) = \text{Newt}(A) + \text{Newt}(g) = Z + \text{Newt}(g).$$

From $\text{supp}(f) \subseteq S$ we get $\text{Newt}(f) \subseteq \text{Conv}(S)$, so for every $r \in \text{supp}(g)$,

$$r + Z \subseteq \text{Newt}(g) + Z = \text{Newt}(f) \subseteq \text{Conv}(S),$$

that is, $r \in R_Z(S)$. Hence $\text{supp}(g) \subseteq R_Z(S)$.

The map $\mathcal{R}_{\mathbb{C}} \rightarrow \{g : \text{supp}(g) \subseteq R_Z(S)\}$, $(c_0, f) \mapsto g = f/A$, is $\mathbb{C}$ -linear and injective (c_0 is determined by f , $A \neq 0$, and division in an integral domain is unique), so

$$\dim_{\mathbb{Q}} \mathcal{R} = \dim_{\mathbb{C}} \mathcal{R}_{\mathbb{C}} \leq |R_Z(S)|. \quad \square$$

Remark 2.6. By lattice-convexity, $r \in R_Z(S)$ implies $(r + Z) \cap \mathbb{Z}^2 \subseteq \text{Conv}(S) \cap \mathbb{Z}^2 = S$. If $R_Z(S) = \emptyset$, then Lemma 2.5 gives $\dim U_S \geq |S| + 1$; but U_S is a space of functions on the finite set $\mathcal{L}$, so $P_{\theta}(S) = |\mathcal{L}| \geq \dim U_S \geq |S| + 1$ and the main theorem holds. This case covers point and segment windows, since Z has non-empty interior ($m \geq 2$ and $v_1 \nparallel v_2$). From now on we assume $R_Z(S) \neq \emptyset$.

3. A APPLIED TO THE COLOUR FIELD YIELDS A DOUBLY PERIODIC BACKGROUND

3.1. Realised sectors and adjacent sectors. The complement $\mathbb{R}^2 \setminus \bigcup_i \mathbb{R}v_i$ has exactly $2m$ connected components, open cones: order the $2m$ rays $\{\sigma v_i : \sigma = \pm, i \in [m]\}$ by angle; the components are the *sectors* between consecutive rays. On each sector every π_i has constant sign; write $\epsilon(K) \in \{+, -\}^m$ for the sign vector of the sector K , and call these $2m$ sign vectors *realised*. (For $m \geq 3$ the remaining $2^m - 2m$ sign vectors are not realised.) For a realised ϵ put

$$\Theta_{\epsilon} := \sum_i \epsilon_i^{\epsilon_i}, \quad \epsilon_i^+ = R_i, \quad \epsilon_i^- = L_i,$$

which is doubly periodic and preserved by every H_j , as is its colour field $e_{\Theta_{\epsilon}} = (\mathbb{K}[\Theta_{\epsilon} = a])_a$.

Lemma 3.0 (Sectors adjacent along a ray). *Let the ray σv_i be the common boundary ray of two sectors K, K' . Then $\epsilon(K)$ and $\epsilon(K')$ differ exactly in the coordinate i , and for $j \neq i$ one has $\epsilon(K)_j = \epsilon(K')_j = \text{sign}(\sigma \pi_j(v_i))$. Consequently*

$$\{\Theta_{\epsilon(K)}, \Theta_{\epsilon(K')}\} = \{G_i^{\sigma, L}, G_i^{\sigma, R}\}.$$

In the angular order, the $2m$ sectors form a $2m$ -cycle under the relation of sharing a boundary ray.

Proof. The ray σv_i lies on $\mathbb{R}v_i$ and on no $\mathbb{R}v_j$ with $j \neq i$, the directions being pairwise non-parallel. A small circular arc centred at σv_i crosses only $\mathbb{R}v_i$, so the sign vectors of the two sectors on either side differ only in coordinate i ; for $j \neq i$, the sign of π_j is constant on the arc and equals $\text{sign} \pi_j(\sigma v_i) = \text{sign}(\sigma \pi_j(v_i))$. By (2.0), $\epsilon_j^{\epsilon(K)_j} = \epsilon_j^{\sigma}$ is precisely the tail choice made in B_i^{σ} , so $\Theta_{\epsilon(K)}$ and $\Theta_{\epsilon(K')}$ are $R_i + B_i^{\sigma}$ and $L_i + B_i^{\sigma}$ in the order dictated by ϵ_i . The cycle structure comes from the angular ordering. $\square$

Lemma 3.1. *For every ray σv_i ,*

$$A(T) e_{G_i^{\sigma, R}} = A(T) e_{G_i^{\sigma, L}}.$$

Consequently $A(T) e_{\Theta_{\epsilon}}$ takes one and the same value for all realised ϵ ; write

$$b = (b_a)_{a \in \mathbb{F}_p} := A(T) e_{\Theta_{\epsilon}}.$$

This b is doubly periodic and preserved by every H_j .

Proof. Colour by colour,

$$\mathbb{K}[G_i^{\sigma, R} = a] - \mathbb{K}[G_i^{\sigma, L} = a] = d_{i, \sigma, L, a} - d_{i, \sigma, R, a},$$

which is killed by $A_i(T^{v_i})$ by (P3), hence by A . By Lemma 3.0 the backgrounds of two sectors adjacent along a ray are exactly $G_i^{\sigma, R}$ and $G_i^{\sigma, L}$, so their values of Ae agree; and the $2m$ sectors form a cycle under this adjacency, hence a connected graph, so $Ae_{\Theta_{\epsilon}}$ is the same for all realised

sectors. Double periodicity and H_j -invariance follow from the corresponding properties of Θ_ϵ together with the fact that A commutes with translations. $\square$

Remark 3.1' (The lemma does not extend to unrealised tail choices). The generalisation “ $Ae_{\Theta_\epsilon} = Ae_{\Theta_{\epsilon'}}$, for all $\epsilon, \epsilon' \in \{L, R\}^m$, realised or not” is *false*. The reason is that Λ_i is defined using only the backgrounds $B_i^\pm$, whereas a mixed tail choice B may have periods along v_i that $B_i^\pm$ do not have, so the spectrum of $\mathcal{K}[R_i + B = a] - \mathcal{K}[L_i + B = a]$ can fall outside Λ_i . An explicit counterexample is in Appendix C.3. Lemma 3.2 uses only realised sectors, so the main line is unaffected.

3.2. $Ae_\theta - b$ has finite support.

Lemma 3.2. *$Ae_\theta - b$ has finite support.*

Proof. Write $M_A = \text{supp}(A)$. Choose $c_i \geq \max(|\ell_i|, |r_i|)$ so that $\{\pi_i < -c_i\} \subseteq \{\pi_i < \ell_i\}$ and $\{\pi_i > c_i\} \subseteq \{\pi_i > r_i\}$, and put

$$\rho_i = c_i + \max_{s \in M_A} |\pi_i(s)|, \quad \Sigma_i = \{z : |\pi_i(z)| \leq \rho_i\}.$$

Call i *active at z* if $z \in \Sigma_i$; otherwise all of $z + M_A$ lies in one and the same tail region of component i , the one on the side $\text{sign } \pi_i(z)$.

(a) *No active direction.* Then $|\pi_i(z)| > \rho_i > 0$ for every i ; put $\epsilon_i = \text{sign } \pi_i(z)$. Regarded as a point of $\mathbb{R}^2$, z lies on no $\mathbb{R}v_i$ and has sign vector ϵ , so ϵ is realised; and $\theta = \Theta_\epsilon$ on $z + M_A$. Hence $(Ae_\theta)(z) = (Ae_{\Theta_\epsilon})(z) = b(z)$.

(b) *At least two active directions.* Then $z \in \Sigma_i \cap \Sigma_j$ with $i \neq j$, an intersection of two strips of bounded width in non-parallel directions, which is bounded and so contains only finitely many lattice points.

(c) *Exactly one active direction i .* Write $z = tv_i + qu_i$ in the basis (v_i, u_i) , so $\pi_i(z) = q$ and $|q| \leq \rho_i$. For $j \neq i$,

$$\pi_j(z) = t \pi_j(v_i) + q \pi_j(u_i), \quad \pi_j(v_i) = \det(v_j, v_i) \neq 0.$$

If

$$|t| > \frac{\rho_i \max_{j \neq i} |\pi_j(u_i)| + \max_{j \neq i} \rho_j}{\min_{j \neq i} |\pi_j(v_i)|}, \quad (3.1)$$

then for every $j \neq i$ and every $s \in M_A$,

$$|\pi_j(z + s)| \geq |t| |\pi_j(v_i)| - |q| |\pi_j(u_i)| - |\pi_j(s)| > \rho_j - |\pi_j(s)| \geq \rho_j - \max_{s' \in M_A} |\pi_j(s')| = c_j,$$

and, because $|t| |\pi_j(v_i)| > |q| |\pi_j(u_i)| + |\pi_j(s)|$, also $\text{sign } \pi_j(z + s) = \text{sign}(t \pi_j(v_i))$. So all of $z + M_A$ lies in the tail region of component j determined by $\text{sign}(t \pi_j(v_i))$. By (2.0) these are exactly the tail choices made in $B_i^{\text{sign}(t)}$, so on $z + M_A$

$$\theta = F_i + B_i^{\text{sign}(t)} = \Xi_i^\sigma, \quad \sigma = \text{sign}(t),$$

whence $(Ae_\theta)(z) = (Ae_{\Xi_i^\sigma})(z)$. By (P3), $A_i(T^{v_i})$ kills $e_{\Xi_i^\sigma} - e_{G_i^{\sigma, R}} = (d_{i, \sigma, R, a})_a$, so

$$(Ae_{\Xi_i^\sigma})(z) = (Ae_{G_i^{\sigma, R}})(z) = b(z),$$

the last step because $G_i^{\sigma, R}$ is the background of one of the realised sectors adjacent along the ray σv_i (Lemma 3.0).

Only finitely many t violate (3.1), and these combine with $|q| \leq \rho_i$ to give finitely many lattice points. Combining (a), (b), (c), $\text{supp}(Ae_\theta - b)$ is finite. $\square$

3.3. Conclusion.

Proposition 3.3. $A(T)e_\theta = b$. In particular

$$A\eta = b_\eta := \sum_a w(a)b_a$$

is a doubly periodic field preserved by every H_i .

Proof. By (1.1), $De_\theta = 0$; and b is preserved by every H_i , so $Db = 0$. Hence $D(Ae_\theta - b) = ADe_\theta - Db = 0$. By Lemma 3.2 this function has finite support, so Lemma 1.2 (with $D \neq 0$) forces it to vanish. $\square$

4. AT LEAST ONE TWO-POINT DIFFERENCE IS NON-ZERO

Put $Q_i = \prod_{j \neq i} (T^{H_j} - 1)$ as in (0.2), and $f_i = Q_i \eta$. From $D = (T^{H_i} - 1)Q_i$ and $D\eta = 0$ it follows that f_i has period H_i .

Lemma 4.1. *The support of f_i is contained in a strip of bounded width in the direction v_i , and $f_i \neq 0$.*

Proof. (a) *A global identity.* We have $f_i = T^{nH_i} f_i$ for every n . Let $n \rightarrow +\infty$: on every fixed finite set $T^{nH_i} \theta \rightarrow \Xi_i^+$, the convergence being eventual pointwise stabilisation, so $T^{nH_i} \eta \rightarrow w(\Xi_i^+)$ and therefore

$$f_i = Q_i w(\Xi_i^+). \quad (4.1)$$

Each factor $T^{H_j} - 1$ with $j \neq i$ annihilates $w(G_i^{+,R})$, since H_j preserves that pure-tail field, so

$$f_i = Q_i \delta_i, \quad \delta_i := w(\Xi_i^+) - w(G_i^{+,R}). \quad (4.2)$$

Using $G_i^{+,L}$ instead gives $f_i = Q_i \delta'_i$ with $\delta'_i = w(\Xi_i^+) - w(G_i^{+,L})$.

(b) *A strip of bounded width.* By (P1), $\delta_i \equiv 0$ on $\{\pi_i > r_i\}$ and $\delta'_i \equiv 0$ on $\{\pi_i < \ell_i\}$. Put $c = \max_{s \in \mathcal{E}_i} |\pi_i(s)|$. By (4.2) and (0.2), $f_i(z) \neq 0$ forces $\delta_i(z+s) \neq 0$ for some $s \in \mathcal{E}_i$, whence $\pi_i(z) \leq r_i + c$; the other identity gives $\pi_i(z) \geq \ell_i - c$. So $\text{supp}(f_i) \subseteq \{\ell_i - c \leq \pi_i \leq r_i + c\}$.

(c) *Non-vanishing.* First, $\delta_i \neq 0$: otherwise $w(\Xi_i^+) = w(G_i^{+,R})$, and injectivity of w gives $\Xi_i^+ = G_i^{+,R}$; subtracting B_i^+ gives $F_i = R_i$, which is doubly periodic, contradicting (S2).

Since δ_i vanishes on $\{\pi_i > r_i\}$ but is not identically zero,

$$\pi^* = \max\{\pi_i(z) : \delta_i(z) \neq 0\} (\leq r_i)$$

exists. Choose z^* with $\delta_i(z^*) \neq 0$ and $\pi_i(z^*) = \pi^*$.

Expand Q_i formally as in (0.2). We have $\pi_i(H_C) = \sum_{j \in C} \pi_i(H_j)$ with every $\pi_i(H_j) \neq 0$ for $j \neq i$. Put

$$C_{\min} = \{j \neq i : \pi_i(H_j) < 0\}.$$

For any $C \subseteq [m] \setminus \{i\}$ with $C \neq C_{\min}$,

$$\pi_i(H_C) - \pi_i(H_{C_{\min}}) = \sum_{j \in C \setminus C_{\min}} \pi_i(H_j) - \sum_{j \in C_{\min} \setminus C} \pi_i(H_j) > 0,$$

because the terms of the first sum are positive, those of the second are negative, and at least one of the two sums is non-empty. So $\pi_i(H_{C_{\min}})$ is the *unique* minimum. This is a statement about the π_i -projections and is unaffected by any coincidences among the points H_C themselves.

Evaluate at $z = z^* - H_{C_{\min}}$:

$$f_i(z) = \sum_C (-1)^{m-1-|C|} \delta_i(z + H_C).$$

The term $C = C_{\min}$ contributes $\pm\delta_i(z^*) \neq 0$; for $C \neq C_{\min}$,

$$\pi_i(z + H_C) = \pi^* + \left(\pi_i(H_C) - \pi_i(H_{C_{\min}}) \right) > \pi^*,$$

so δ_i vanishes there. Hence $f_i(z) = \pm\delta_i(z^*) \neq 0$. $\square$

Lemma 4.2. *For every d the function $J(d, \cdot) := D_z[\eta(z)\eta(z+d)]$ has finite support; and there exists d with $J(d, \cdot) \neq 0$.*

Proof. Finite support: $z \mapsto \eta(z)\eta(z+d)$ is a local function of θ , with window $W = \{0, d\}$; apply Lemma 1.1.

Non-vanishing: choose $i \neq j$, possible since $m \geq 2$. By Lemma 4.1 choose x_i, x_j with $f_i(x_i) \neq 0$ and $f_j(x_j) \neq 0$. Put $d_0 = x_j - x_i$ and

$$C(z) = f_i(z) f_j(z + d_0).$$

Then $C(x_i) = f_i(x_i)f_j(x_j) \neq 0$. Now $\text{supp}(f_i)$ lies in a strip of bounded width in the direction v_i and $\text{supp}(f_j(\cdot + d_0))$ in one of direction v_j , with $v_i \not\parallel v_j$, so $\text{supp}(C)$ is bounded. By Lemma 1.2, $DC \neq 0$.

On the other hand, expanding formally by (0.2),

$$C(z) = \left(\sum_{C_1} \pm \eta(z + H_{C_1}) \right) \left(\sum_{C_2} \pm \eta(z + d_0 + H_{C_2}) \right) = \sum_{C_1, C_2} \pm \eta(z + u_1) \eta(z + u_2),$$

with $u_1 = H_{C_1}$ and $u_2 = d_0 + H_{C_2}$. By translation invariance of D ,

$$DC = \sum_{C_1, C_2} \pm T^{u_1} \left(D_z[\eta(z)\eta(z + u_2 - u_1)] \right) = \sum_{C_1, C_2} \pm T^{u_1} J(u_2 - u_1, \cdot).$$

If every $J(d, \cdot)$ vanished, then $DC = 0$, a contradiction. $\square$

5. THE BI-RECURSION AND THE QUOTIENT ALGEBRA

5.1. The bi-recursion. Regard J as a function of the pair (d, z) : $J(d, z) = D_z[\eta(z)\eta(z+d)]$. Write $T_d^s J(d, z) = J(d+s, z)$ and $T_z^s J(d, z) = J(d, z+s)$.

Lemma 5.1.

$$A(T_d) J = 0, \quad A(T_z T_d^{-1}) J = 0. \quad (5.1)$$

Proof. First identity. For any Laurent polynomial $g = \sum_s g_s T^s$,

$$g(T_d) J(d, z) = \sum_s g_s D_z[\eta(z)\eta(z+d+s)] = D_z[\eta(z)(g(T)\eta)(z+d)].$$

Take $g = A$; by Proposition 3.3, $(A\eta)(z+d) = b_\eta(z+d)$. Expanding D_z by (0.1),

$$D_z[\eta(z)b_\eta(z+d)] = \sum_C (-1)^{m-|C|} \eta(z + H_C) b_\eta(z + H_C + d).$$

Each H_C is a sum of vectors H_i and so preserves b_η ; therefore $b_\eta(z + H_C + d) = b_\eta(z + d)$ and

$$= b_\eta(z + d) \sum_C (-1)^{m-|C|} \eta(z + H_C) = b_\eta(z + d) (D\eta)(z) = 0.$$

Second identity. The operator $(T_z T_d^{-1})^s$ sends (d, z) to $(d-s, z+s)$: the first observation point moves, $z \mapsto z+s$, while the second, $z+d$, does not. Hence

$$g(T_z T_d^{-1}) J(d, z) = \sum_s g_s D_{z'}[\eta(z')\eta(z' + d - s)] \Big|_{z'=z+s}$$

$$\begin{aligned}
&= \sum_C (-1)^{m-|C|} \sum_s g_s \eta(z + s + H_C) \eta(z + H_C + d) \\
&= \sum_C (-1)^{m-|C|} (g(T)\eta)(z + H_C) \eta(z + H_C + d).
\end{aligned}$$

Take $g = A$: then $(A\eta)(z + H_C) = b_\eta(z + H_C) = b_\eta(z)$, so the sum equals

$$b_\eta(z) \sum_C (-1)^{m-|C|} \eta(z + H_C + d) = b_\eta(z) \cdot (D\eta)(z + d) = 0. \quad \square$$

5.2. The quotient algebra. Let $X = (X_1, X_2)$ be *independent indeterminates*, $K = \mathbb{C}(X_1, X_2)$, and $R = K[Y_1^{\pm 1}, Y_2^{\pm 1}]$. For $u \in \mathbb{Z}^2$ write $Y^u = Y_1^{u_1} Y_2^{u_2}$ and $X^u = X_1^{u_1} X_2^{u_2}$; the monomials $\{Y^u\}_{u \in \mathbb{Z}^2}$ are a K -basis of R . Put

$$a_i = A_i(Y^{v_i}), \quad c_i = A_i(X^{v_i} Y^{-v_i}), \quad a = \prod_i a_i = A(Y), \quad c = \prod_i c_i = A(X/Y).$$

Lemma 5.2. *As a K -vector space, $R/(a, c)$ is spanned by the images of $\{Y^d : d \in (Z - Z) \cap \mathbb{Z}^2\}$.*

Proof. Step 1: no common zeros, and the Chinese remainder theorem. Extend to $\overline{K}$ and test in $(\overline{K}^\times)^2$. Note that $\mathbb{C}$ is algebraically closed, so an element of $\overline{K}$ that is algebraic over $\mathbb{C}$ lies in $\mathbb{C}$, whereas X^u with $u \neq 0$ is transcendental over $\mathbb{C}$.

(i) a_i and c_i have no common zero. From $a_i = 0$ we get $Y^{v_i} = \lambda \in \Lambda_i$; from $c_i = 0$ we get $X^{v_i} Y^{-v_i} = \mu \in \Lambda_i$. Multiplying, $X^{v_i} = \lambda \mu \in \mathbb{C}$, a contradiction.

(ii) If $a_i = a_k = 0$ with $i \neq k$, then every $c_j \neq 0$. From $Y^{v_i} = \lambda$, $Y^{v_k} = \lambda'$ and $\Delta = \det(v_i, v_k) \neq 0$, the elements Y_1^Δ and Y_2^Δ are products of integer powers of λ, λ' , so Y_1, Y_2 are algebraic over $\mathbb{C}$ and hence lie in $\mathbb{C}^\times$. If some $c_j = 0$ then $X^{v_j} = \mu Y^{v_j} \in \mathbb{C}$, a contradiction.

(iii) If $c_j = c_l = 0$ with $j \neq l$, then every $a_i \neq 0$. Put $W = X/Y$, that is $W_1 = X_1/Y_1$ and $W_2 = X_2/Y_2$. From $W^{v_j} = \mu$ and $W^{v_l} = \mu'$ we get, as in (ii), $W_1, W_2 \in \mathbb{C}^\times$, that is $Y_1 = \alpha X_1$ and $Y_2 = \beta X_2$ with $\alpha, \beta \in \mathbb{C}^\times$. If some $a_i = 0$ then $Y^{v_i} = \lambda$, that is $\alpha^{v_{i1}} \beta^{v_{i2}} X^{v_i} = \lambda$ and $X^{v_i} \in \mathbb{C}$, a contradiction.

By (ii), for $i \neq k$ the ideal (a_i, a_k, c) has no zero in $(\overline{K}^\times)^2$, so it equals (1) in $\overline{K}[Y^\pm]$ by the Nullstellensatz, and descends to R by faithful flatness of $R \rightarrow \overline{K}[Y^\pm]$: $(a_i, a_k, c) = R$. That is, the a_i are pairwise coprime in $R/(c)$. By the Chinese remainder theorem,

$$R/(a, c) = R/(\prod_i a_i, c) \simeq \prod_i R/(a_i, c).$$

In $R/(a_i)$: by (i), $(a_i, c_i) = R$, so c_i is a unit; by (iii), for $j \neq l$ with $j, l \neq i$ one has $(a_i, c_j, c_l) = R$, so the c_j are pairwise coprime in $R/(a_i)$. A second application of the Chinese remainder theorem gives

$$R/(a_i, c) = R/(a_i, \prod_j c_j) \simeq \prod_{j \neq i} R/(a_i, c_j),$$

and combining, with the isomorphism given by the natural projections,

$$R/(a, c) \simeq \prod_{i \neq j} R/(a_i, c_j). \quad (5.2)$$

Step 2: a monomial spanning set for each factor. Fix $i \neq j$. Put $u = Y^{v_i}$ and $w = Y^{-v_j}$, let $\mathfrak{L} = \mathbb{Z}v_i + \mathbb{Z}(-v_j)$ and $\nu = [\mathbb{Z}^2 : \mathfrak{L}] = |\det(v_i, v_j)|$. Take the ν lattice points $r_1, \dots, r_\nu$ of the half-open fundamental domain

$$\Pi = \{\alpha v_i - \beta v_j : 0 \leq \alpha < 1, 0 \leq \beta < 1\},$$

a complete set of representatives for $\mathbb{Z}^2/\mathfrak{L}$. Then R is free as a $K[u^\pm, w^\pm]$ -module with basis $\{Y^{r_n}\}$.

The ideal (a_i, c_j) is generated by the elements $a_i = A_i(\mathbf{u})$ and $c_j = A_j(X^{v_j}\mathbf{w})$ of the subring $K[\mathbf{u}^\pm, \mathbf{w}^\pm]$, so

$$R/(a_i, c_j) \simeq \bigoplus_{n=1}^{\nu} Y^{r_n} \cdot K[\mathbf{u}^\pm, \mathbf{w}^\pm]/(A_i(\mathbf{u}), A_j(X^{v_j}\mathbf{w})).$$

Now $A_i(\mathbf{u})$ has degree d_i in $\mathbf{u}$ with both extreme coefficients non-zero (A_i is monic and $A_i(0) \neq 0$), so any integer power of $\mathbf{u}$ reduces into the K -span of $\{\mathbf{u}^0, \dots, \mathbf{u}^{d_i-1}\}$; and $A_j(X^{v_j}\mathbf{w})$ has degree d_j in $\mathbf{w}$ with leading coefficient $X^{d_j v_j} \neq 0$ and constant term $A_j(0) \neq 0$, so any integer power of $\mathbf{w}$ reduces into the span of $\{\mathbf{w}^0, \dots, \mathbf{w}^{d_j-1}\}$. Hence

$$\left\{ Y^{r_n + \alpha v_i - \beta v_j} : 1 \leq n \leq \nu, 0 \leq \alpha < d_i, 0 \leq \beta < d_j \right\} \quad (5.3)$$

spans $R/(a_i, c_j)$. Since $r_n \in \Pi$ — its v_i -coefficient and its $(-v_j)$ -coefficient both lie in $[0, 1]$ — all the exponents in (5.3) lie in

$$P_{ij} := [0, d_i v_i] + [0, -d_j v_j]. \quad (5.4)$$

Step 3: projection onto a factor. Put

$$E_{ij} = \prod_{k \neq i} a_k \prod_{l \neq j} c_l.$$

In the decomposition (5.2):

- on a factor (i', j') with $i' \neq i$: the factor $a_{i'}$ occurs in E_{ij} and is $\equiv 0$ there, so $E_{ij} \equiv 0$;
- on a factor (i, j') with $j' \neq j$: the factor $c_{j'}$ occurs and is $\equiv 0$, so $E_{ij} \equiv 0$;
- on the factor (i, j) : by (i), (ii), (iii), the ideals (a_i, c_j, a_k) for $k \neq i$ and (a_i, c_j, c_l) for $l \neq j$ have no zero in $(\overline{K}^\times)^2$, hence equal R by the Nullstellensatz and faithfully flat descent. Now $(a_i, c_j, x) = R$ means precisely that there is $w \in R$ with $wx \equiv 1 \pmod{(a_i, c_j)}$, so each a_k and each c_l is a unit in $R/(a_i, c_j)$, and so is their product E_{ij} .

Therefore the image of $E_{ij} \cdot (5.3)$ spans the factor (i, j) and vanishes on all the others. Every monomial exponent occurring in $E_{ij} \cdot Y^e$ lies in $\text{Newt}(E_{ij}) + e$, and

$$\begin{aligned} \text{Newt}(E_{ij}) + P_{ij} &\subseteq \sum_{k \neq i} [0, d_k v_k] + \sum_{l \neq j} [0, -d_l v_l] + P_{ij} \\ &= \sum_k [0, d_k v_k] + \sum_l [0, -d_l v_l] = Z + (-Z) = Z - Z, \end{aligned}$$

using $\text{Newt}(a_k) = [0, d_k v_k]$, $\text{Newt}(c_l) = [0, -d_l v_l]$ and the fact that the Newton polygon of a product is the Minkowski sum. Taking the union over all $i \neq j$ gives the spanning statement. $\square$

5.3. The witness lies in $Z - Z$.

Proposition 5.3. *There is $d \in (Z - Z) \cap \mathbb{Z}^2$ with $d \neq 0$ and $J(d, \cdot) \neq 0$.*

Proof. By Lemma 4.2 each $J(d, \cdot)$ has finite support, so

$$\hat{J}(d; X) = \sum_z J(d, z) X^{-z} \in \mathbb{C}[X_1^\pm, X_2^\pm] \subset K$$

is well defined, the supports for different d being allowed to differ. Define a K -linear functional $L: R \rightarrow K$ by setting $L(Y^d) = \hat{J}(d; X)$ on the K -basis and extending K -linearly. That L is K -linear, and not merely $\mathbb{C}$ -linear, is what makes the next step work: coefficients $X^s \in K$ may be pulled out of L .

L kills the ideal (a, c) . Write $A = \sum_s A_s T^s$, so that $a = A(Y) = \sum_s A_s Y^s$ and $c = A(X/Y) = \sum_s A_s X^s Y^{-s}$.

By the first identity of (5.1), for every (d, z) we have $\sum_s A_s J(d + s, z) = 0$. Taking the Laurent transform in z ,

$$\sum_s A_s \hat{J}(d + s; X) = 0 \implies L(Y^d \cdot a) = \sum_s A_s L(Y^{d+s}) = 0.$$

By the second identity of (5.1), for every (d, z) we have $\sum_s A_s J(d - s, z + s) = 0$. Taking the Laurent transform in z and noting

$$\sum_z J(d - s, z + s) X^{-z} = \sum_{z'} J(d - s, z') X^{-z'+s} = X^s \hat{J}(d - s; X),$$

we get $\sum_s A_s X^s \hat{J}(d - s; X) = 0$, that is $L(Y^d \cdot c) = \sum_s A_s X^s L(Y^{d-s}) = 0$.

For an arbitrary $g = \sum_d g_d Y^d \in R$ with $g_d \in K$, K -linearity gives $L(ga) = \sum_d g_d L(Y^d a) = 0$, and likewise $L(gc) = 0$. So L vanishes on (a, c) and descends to $\bar{L}: R/(a, c) \rightarrow K$.

Conclusion. If $J(d, \cdot) = 0$ for every $d \in (Z - Z) \cap \mathbb{Z}^2$, then $\bar{L}$ vanishes on a spanning set supplied by Lemma 5.2, hence $\bar{L} = 0$, hence $L = 0$, hence $\hat{J}(d; X) = 0$ for every d , that is $J \equiv 0$ — contradicting Lemma 4.2.

$d \neq 0$. We have $J(0, z) = D_z[\eta(z)^2]$, and $\eta^2 = w'(\theta)$ with $w'(a) = w(a)^2$ is a single-site colour function, so $D\eta^2 = 0$ by (1.1). $\square$

6. PUTTING TWO POINTS INTO ONE INTEGRAL ZONOTOPE

Lemma 6.1. *Let $Z = \sum_{i=1}^m [0, d_i v_i]$ with integers $d_i \geq 1$, $v_i \in \mathbb{Z}^2$, $m \geq 2$ and $v_1 \nparallel v_2$. Then*

$$(Z - Z) \cap \mathbb{Z}^2 = (Z \cap \mathbb{Z}^2) - (Z \cap \mathbb{Z}^2). \quad (6.1)$$

Proof. The inclusion “ $\supseteq$ ” is clear.

“ $\subseteq$ ”: put $c_0 = \sum_i d_i v_i \in \mathbb{Z}^2$. For $x = \sum_i x_i \in Z$ with $x_i \in [0, d_i v_i]$ we have $c_0 - x = \sum_i (d_i v_i - x_i)$ with $d_i v_i - x_i \in [0, d_i v_i]$, so $c_0 - x \in Z$; that is, $c_0 - Z \subseteq Z$. Applying this once more, $Z = c_0 - (c_0 - Z) \subseteq c_0 - Z$, so $c_0 - Z = Z$ and $-Z = Z - c_0$. Hence

$$Z - Z = Z + (-Z) = Z + Z - c_0 = 2Z - c_0.$$

Let $d \in (Z - Z) \cap \mathbb{Z}^2$ and put $x = d + c_0 \in 2Z \cap \mathbb{Z}^2$.

The vertices of Z are subset sums of the vectors $d_i v_i$ and so are lattice points; thus Z is a two-dimensional lattice polygon. Take a *unimodular triangulation* of Z : every two-dimensional lattice polygon can be subdivided into lattice triangles of area $\frac{1}{2}$ whose vertices are lattice points of Z . Since $x/2 \in Z$, it lies in some unimodular triangle $\triangle ABC$ (boundary included), and $(B - A, C - A)$ is a basis of $\mathbb{Z}^2$. Write

$$\frac{1}{2}x = A + \alpha(B - A) + \beta(C - A), \quad \alpha, \beta \geq 0, \alpha + \beta \leq 1.$$

Then $x - 2A = (2\alpha)(B - A) + (2\beta)(C - A)$. The left-hand side is a lattice point and $(B - A, C - A)$ is a lattice basis, so $2\alpha, 2\beta \in \mathbb{Z}_{\geq 0}$ with $2\alpha + 2\beta \leq 2$. The cases are:

$(2\alpha, 2\beta)$	x	decomposition
(0, 0)	$2A$	$A + A$
(1, 0)	$2A + (B - A) = A + B$	$A + B$
(0, 1)	$A + C$	$A + C$
(2, 0)	$2A + 2(B - A) = 2B$	$B + B$
(0, 2)	$2C$	$C + C$
(1, 1)	$2A + (B - A) + (C - A) = B + C$	$B + C$

In every case $x = p + q$ with $p, q \in \{A, B, C\} \subseteq Z \cap \mathbb{Z}^2$.

Finally $d = x - c_0 = p - (c_0 - q)$, and $c_0 - q \in c_0 - Z = Z$ is a lattice point. $\square$

Corollary 6.2. *The vector d of Proposition 5.3 can be written $d = q_1 - q_0$ with $q_0, q_1 \in Z \cap \mathbb{Z}^2$. Putting $q = q_0$, both q and $q + d$ lie in $Z \cap \mathbb{Z}^2$.*

7. QUADRATIC CHARACTERS MAKE UP THE WHOLE LINEAR DEFICIT

Fix d and q as in Proposition 5.3 and Corollary 6.2, and put

$$h(z) = \eta(z) \eta(z + d), \quad J := Dh \neq 0 \text{ of finite support.}$$

Lemma 7.1. *For every $r \in R_Z(S)$ one has $r + q \in S$ and $r + q + d \in S$.*

Proof. We have $r + Z \subseteq \text{Conv}(S)$ and $q, q + d \in Z \cap \mathbb{Z}^2$, so

$$r + q, r + q + d \in (r + Z) \cap \mathbb{Z}^2 \subseteq \text{Conv}(S) \cap \mathbb{Z}^2 = S. \quad \square$$

Consequently, for each $r \in R_Z(S)$ the *quadratic scalar character*

$$\Phi_r: \mathcal{L} \rightarrow \mathbb{Q}, \quad \Phi_r(\gamma) = w(\gamma(r + q)) \cdot w(\gamma(r + q + d))$$

is well defined, and $\Phi_r(\theta|_{u+S}) = \eta(u + r + q) \eta(u + r + q + d) = h(u + r + q)$.

Lemma 7.2. *The family $\{\Phi_r\}_{r \in R_Z(S)}$ is linearly independent in $\{\mathcal{L} \rightarrow \mathbb{Q}\}/U_S$.*

Proof. Suppose not: there are (c_r) , not all zero, and $\psi \in U_S$ with $\sum_r c_r \Phi_r = \psi$ on $\mathcal{L}$. Substituting $\gamma = \theta|_{u+S}$,

$$\sum_{r \in R_Z(S)} c_r h(u + r + q) = c_0 + \sum_{s \in S} \mu_s \eta(u + s) \quad \text{for all } u.$$

Apply D in u to both sides. The right-hand side gives $Dc_0 + \sum_s \mu_s T^s(D\eta) = 0$ by (1.1); the left-hand side gives $(\sum_r c_r T^{r+q})(Dh) = (\sum_r c_r T^{r+q})J$.

So $(\sum_r c_r T^{r+q})J = 0$. The exponents $r + q$ for $r \in R_Z(S)$ are distinct and the (c_r) are not all zero, so that Laurent polynomial is non-zero; and J is non-zero of finite support. This contradicts Lemma 1.2. $\square$

Theorem 7.3 (Theorem T). $P_\theta(S) \geq |S| + 1$.

Proof. The case $R_Z(S) = \emptyset$ is Remark 2.6. Assume $R_Z(S) \neq \emptyset$ and put

$$V = U_S + \text{span}_{\mathbb{Q}}\{\Phi_r : r \in R_Z(S)\} \subseteq \{\mathcal{L} \rightarrow \mathbb{Q}\}.$$

By Lemma 7.2, $\dim V = \dim U_S + |R_Z(S)|$; by Lemma 2.5,

$$\dim V \geq (|S| - |R_Z(S)| + 1) + |R_Z(S)| = |S| + 1.$$

On the other hand V is a space of functions on the finite set $\mathcal{L}$, so $\dim V \leq |\mathcal{L}| = P_\theta(S)$. $\square$

Remark 7.4 (The balance). The zonotope Z enters the proof twice and with opposite signs: it is the *cost* of the affine relations, Lemma 2.5 subtracting $|R_Z(S)|$, and it is the *capacity* of the quadratic witness, Lemma 7.1 supplying exactly $|R_Z(S)|$ positions. The two cancel exactly, independently of the shape of S and of the size of Z . The numerical experiments of Appendix C show that Lemma 2.5 is an equality on every instance of Case B that we generated.

8. REDUCTION TO THE CONVEX NIVAT CONJECTURE

This section starts from a non-periodic configuration ξ of low convex complexity, extracts a star configuration from X_ξ , and contradicts Theorem 7.3. The chain is

$$\begin{aligned} \text{Theorem 8.4 [8]} &\longrightarrow \text{Lemma 8.5 (mod } p) \longrightarrow \text{Theorems 8.6 and 8.7 [1]} \\ &\longrightarrow \text{Proposition 8.9 (first half-plane, this paper)} \\ &\longrightarrow \text{Assumption 8.12 (second half-plane, [4])} \longrightarrow \text{Lemma 8.14} \longrightarrow \text{Theorem 7.3.} \end{aligned}$$

Apart from Assumption 8.12, every link is either a published result or proved here in full. The dependence on [4] is confined to its Proposition 3.1(ii): its Lemma 2.10, Lemma 2.11, Claim 3.2 and Proposition 3.1(i) are not on the chain, and the modular minimality theorem of [2] enters only where Assumption 8.12 is docked onto the proof given in [4] (Remark 8.13).

Theorem 8.1 (Structure input). *Let ξ be a counterexample of minimal order in the sense of Remark 8.3. Then there are a prime p with $\mathcal{A} \subset \llbracket p \rrbracket$, a non-periodic $\vartheta \in X_\xi$, an $m \geq 2$ and $\mathbb{F}_p$ -valued $G_1, \dots, G_m$ with $\vartheta = \sum_i G_i$ over $\mathbb{F}_p$, each G_i having a non-zero period h_i , the h_i pairwise non-parallel and no component doubly periodic, such that for every i :*

- (i) G_i is fully periodic on a half-plane U_i whose boundary is parallel to h_i ;
- (ii) G_i is fully periodic on a half-plane V_i disjoint from U_i .

Part (i) is proved in §§8.2–8.3, from published results and Proposition 8.9 of this paper. Part (ii) is the content of [4, Proposition 3.1(ii)] and is quoted here as Assumption 8.12 (§8.4); Theorem 8.15 is conditional on it.

8.1. Half-planes, regions and full periodicity.

Definition. A *half-plane* is a set of the form $\mathcal{H} = \{z \in \mathbb{Z}^2 : \langle z, n \rangle > c\}$ or $\mathcal{H} = \{z \in \mathbb{Z}^2 : \langle z, n \rangle \geq c\}$ with $n \in \mathbb{R}^2 \setminus \{0\}$ and $c \in \mathbb{R}$; we call n its normal. For an integer vector g : if $\langle g, n \rangle \geq 0$ then $\mathcal{H} + g \subseteq \mathcal{H}$; if $\langle g, n \rangle > 0$ then $\bigcup_{N \geq 0} (\mathcal{H} - Ng) = \mathbb{Z}^2$.

Call $R \subseteq \mathbb{Z}^2$ a *lattice-convex region* if $R = C \cap \mathbb{Z}^2$ for some closed convex $C \subseteq \mathbb{R}^2$. Writing $C_R := \overline{\text{Conv}}(R)$ for the closed convex hull, one still has $R = C_R \cap \mathbb{Z}^2$, since $R \subseteq C_R \subseteq C$. Let

$$K_R := \{w \in \mathbb{R}^2 : C_R + w \subseteq C_R\}$$

be the recession cone of C_R , a closed convex cone. If $h \in \mathbb{Z}^2$ satisfies $R + h \subseteq R$, then $C_R + h = \overline{\text{Conv}}(R + h) \subseteq C_R$, that is $h \in K_R$. A half-plane is a lattice-convex region, with $K_R = \{w : \langle w, n \rangle \geq 0\}$.

Following Colle and Garibaldi [4, Definition 2.1], say for a non-empty $\mathcal{U} \subseteq \mathbb{Z}^2$ that $G: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ is *fully periodic on $\mathcal{U}$* if there are $\mathbb{R}$ -linearly independent $h, h' \in \mathbb{Z}^2$ with $\mathcal{U} + h \subseteq \mathcal{U}$, $\mathcal{U} + h' \subseteq \mathcal{U}$, $G(z + h) = G(z)$ and $G(z + h') = G(z)$ for all $z \in \mathcal{U}$. The restriction of a doubly periodic field to any half-plane is fully periodic: take two independent vectors of its period lattice pointing into the half-plane. Conversely:

Lemma 8.2 (Global extension). *Let R be a lattice-convex region and let G be fully periodic on R , with vectors h, h' . Then $\dim K_R = 2$ and $g := h + h' \in \text{int } K_R$; for every $z \in \mathbb{Z}^2$ one has $z + Ng \in R$ for all large N ; and there is a doubly periodic field $\tilde{G}$ with $\text{Per}(\tilde{G}) \supseteq \mathbb{Z}h + \mathbb{Z}h'$ and $\tilde{G} = G$ on R .*

Proof. From $R + h \subseteq R$ and $R + h' \subseteq R$ we get $h, h' \in K_R$; they are linearly independent, so the closed sector they span lies in K_R , its interior is non-empty and contains g , whence $g \in \text{int } K_R$ and $\dim K_R = 2$. Pick $x_0 \in C_R$; by the definition of the recession cone $x_0 + K_R \subseteq C_R$. For

any z we have $z + Ng - x_0 = N(g + \frac{z-x_0}{N})$, and for N large the vector in brackets lies in a neighbourhood of g contained in K_R ; hence $z + Ng \in C_R \cap \mathbb{Z}^2 = R$.

Step 1. For $z \in R$ and $k \geq 0$: $z + kh \in R$ and $G(z + kh) = G(z)$, by induction on k ; likewise for h' and for g .

Definition. For $z \in \mathbb{Z}^2$ pick $N \geq 0$ with $z + Ng \in R$ and set $\tilde{G}(z) = G(z + Ng)$. This is well defined: if $N \leq N'$ both work, then $G(z + N'g) = G((z + Ng) + (N' - N)g) = G(z + Ng)$ by Step 1. Taking $N = 0$ shows $\tilde{G} = G$ on R .

Periodicity. Given z , pick N with $z + Ng \in R$; then also $z + h + Ng \in R$, so $\tilde{G}(z + h) = G((z + Ng) + h) = G(z + Ng) = \tilde{G}(z)$ by Step 1. The same for h' . $\square$

Half-planes are the special case: there $K_R = \{w : \langle w, n \rangle \geq 0\}$ and g satisfies $\langle g, n \rangle > 0$.

8.2. The published input and counterexamples of minimal order.

Remark 8.3 (Counterexamples of minimal order). For a non-periodic configuration ξ of low convex complexity with $\mathcal{A} \subset \mathbb{Z}_+$, write $\text{ord}(\xi)$ for the length of a $\mathbb{Z}$ -minimal periodic decomposition of ξ ; by [8] a configuration of low complexity has a non-trivial annihilator and hence a finite periodic decomposition, so ord is finite. If the convex Nivat conjecture has a counterexample, then one may take a counterexample ξ with $\text{ord}(\xi)$ minimal, and for such a ξ every non-periodic $\xi' \in X_\xi$ is again a counterexample, with $\text{ord}(\xi') = \text{ord}(\xi)$.

Proof. Every finite-window pattern of ξ' is a pattern of ξ , so ξ' has low convex complexity; being non-periodic it is a counterexample, and minimality gives $\text{ord}(\xi') \geq \text{ord}(\xi) =: m$. Conversely write $\xi = \sum_{i \leq m} \xi_i$ minimal, with ξ_i of period h_i ; the directions are pairwise distinct, since two components with parallel periods could be merged, contradicting minimality. Then $\varphi = \prod_i (X^{h_i} - 1)$ annihilates ξ ; an annihilation relation is a linear identity on a finite window and therefore passes to the orbit closure, so φ annihilates ξ' , and by [8] the configuration ξ' has a periodic decomposition with at most m terms, that is $\text{ord}(\xi') \leq m$. $\square$

Consequently every result of [1] proved under the standing assumption that the non-periodic configurations of the orbit closure all have the same order — including [1, Theorem 1.10] and the Claims of its §4 — is available for a counterexample ξ of minimal order; it is under that assumption that Theorem 8.7 below quotes [1, Lemma 4.6].

Theorem 8.4 (Kari-Szabados [8]). *Let $\xi : \mathbb{Z}^2 \rightarrow \mathcal{A}$ with $\mathcal{A} \subset \mathbb{Z}$ finite.*

- (a) *If $P_\xi(S) \leq |S|$ for some non-empty finite $S \subset \mathbb{Z}^2$, then $\text{Ann}_{\mathbb{Z}}(\xi) \neq 0$.*
- (b) *If $\text{Ann}_{\mathbb{Z}}(\xi) \neq 0$, then there are pairwise non-parallel $h_1, \dots, h_m \in \mathbb{Z}^2 \setminus \{0\}$ such that $\varphi = \prod_{i=1}^m (X^{h_i} - 1)$ annihilates ξ ; and every $\eta \in \mathbb{Z}^{\mathbb{Z}^2}$ annihilated by such a φ can be written $\eta = \sum_i \eta_i$ with $\eta_i \in \mathbb{Z}^{\mathbb{Z}^2}$ of period h_i (the values possibly unbounded).*

Part (a) is the linear dependence of the S -patterns, the same argument as in the proof of Proposition 1.3; the two assertions of (b) are the decomposition theorem of [8] and the construction inside its proof.

Lemma 8.5 (Passing to the orbit closure and reducing mod p). *Let ξ be as in Theorem 8.4 with $\mathcal{A} \subset \mathbb{Z}_+$, and let $\varphi = \prod_{i=1}^m (X^{h_i} - 1)$ annihilate ξ . Then for every $\vartheta \in X_\xi$ and every prime $p > \max \mathcal{A}$:*

- (a) *φ annihilates ϑ , so $\vartheta = \sum_i \vartheta_i$ with $\vartheta_i \in \mathbb{Z}^{\mathbb{Z}^2}$ of period h_i ;*
- (b) *putting $\bar{\vartheta}_i := \vartheta_i \bmod p : \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ one has $\sum_i \bar{\vartheta}_i = \bar{\vartheta}$, the values of ϑ being read in $\mathbb{F}_p$ (no colours are merged, since $\mathcal{A} \subset \llbracket p \rrbracket$), and each $\bar{\vartheta}_i$ has period h_i ;*

- (c) after merging the components whose periods are parallel one may assume the h_i pairwise non-parallel; and if ϑ is non-periodic, at least two components remain.

Proof. (a) At each z the identity $\varphi\vartheta = 0$ involves only $\vartheta|_{z+\text{supp } \varphi}$, so it is a local function of ϑ ; it vanishes on every translate of ξ , and local functions are continuous for the product topology, hence it vanishes on X_ξ . Now apply Theorem 8.4(b). (b) Reduce pointwise. (c) The sum of two components with parallel periods is again periodic (take a common multiple of the two periods); if only one term were left, ϑ would be periodic. $\square$

Lemma 8.5 needs no minimality of the decomposition; minimality appears only in Assumption 8.12, where it is required in order to dock onto [4].

Theorem 8.6 (Colle [1, Theorem 1.9]). *Let $\mathcal{A} \subset \mathbb{Z}$ and let ξ have a non-trivial annihilator. If for every line ℓ either $-\ell \notin \text{ONED}(\xi)$ or $\ell \notin \text{ONED}(\xi)$, then ξ is fully periodic. In particular, if ξ is non-periodic there is a line ℓ with $\pm\ell \in \text{ONED}(\xi)$.*

Its proof uses the periodic-limit theorem of Kari and Moutot [9] and the Boyle–Lind theorem, both published; we do not reprove it.

Theorem 8.7 (Colle [1, Lemma 4.6], the two cases). *Let ξ be a counterexample of minimal order in the sense of Remark 8.3, so that the equal-order assumption of §4.1 of [1] is in force, and let ℓ be as in Theorem 8.6. Then there are a non-periodic $\vartheta \in X_\xi$ and a lattice-convex region R such that $\vartheta|_R$ is fully periodic. Here R is a (ℓ, ℓ') -region: the set of lattice points of a convex unbounded polygonal region whose boundary consists of two unbounded edges, along ℓ and along ℓ' , both nonexpansive lines for X_ξ , and finitely many bounded edges.*

Only three items of this are used below: ϑ is non-periodic, R is a lattice-convex region, and ϑ is fully periodic on R . The position of the two asymptotic directions of the region (Claims 4.11 and 4.16 of [1], Claim 3.2 of [4]) is not needed: Proposition 8.9(c) shows that it is determined by the decomposition itself. We read the conclusion with “fully periodic” as defined in §8.1, that is as in [4, Definition 2.1]: both periods point into the region.

8.3. The first half-plane. This subsection proves Theorem 8.1(i). The only tools are Lemma 8.2, the difference operators Q_i of (0.2), and the finite-state argument of step (6) of Lemma D.7 in Appendix D.

Lemma 8.8 (One-sided finite-state lemma). *Let $G: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ have period kv with v primitive and $k \geq 1$, let (v, u) be a basis of $\mathbb{Z}^2$, so $\det(v, u) = \pm 1$, and define the row coordinate of $z = sv + tu$ by $t = t(z) = \det(v, u) \det(v, z)$. Let $H_1, \dots, H_r \in \mathbb{Z}^2$ with $r \geq 1$ satisfy $\beta_j := t(H_j) \neq 0$, and put $\beta_- := \sum_{\beta_j < 0} \beta_j$ and $\beta_+ := \sum_{\beta_j > 0} \beta_j$. If*

$$\Delta := \prod_{j=1}^r (T^{H_j} - 1) G$$

vanishes on the half-plane $\{t \geq c\}$, then there is $q \geq 1$ with $G(z + qu) = G(z)$ for all z with $t(z) \geq c + \beta_-$; in particular G is fully periodic on $\{t \geq c + \beta_-\}$, with the periods kv and qu , neither of which leaves that half-plane. If $\Delta \equiv 0$, then G is doubly periodic.

Proof. Write $z = sv + tu$. Let $\mathcal{W} := \{f: \mathbb{Z} \rightarrow \mathbb{F}_p : f(s+k) = f(s)\}$, a set of cardinality p^k , and for each row t let $G_t \in \mathcal{W}$ be $G_t(s) := G(sv + tu)$. Write $H_j = \alpha_j v + \beta_j u$. Then

$$(T^{H_j} G)(sv + tu) = G((s + \alpha_j)v + (t + \beta_j)u) = (\rho^{\alpha_j} G_{t+\beta_j})(s),$$

where $(\rho f)(s) := f(s+1)$ is an invertible linear map of $\mathcal{W}$. Since ρ commutes with the translations in the t direction, for every t

$$\Delta_t = \sum_{C \subseteq [r]} (-1)^{r-|C|} \rho^{\alpha_C} G_{t+\beta_C}, \quad \alpha_C := \sum_{j \in C} \alpha_j, \quad \beta_C := \sum_{j \in C} \beta_j.$$

The maximum $\beta_+ = \max_C \beta_C$ is attained only at $C_+ := \{j : \beta_j > 0\}$ and the minimum $\beta_- = \min_C \beta_C$ only at $C_- := \{j : \beta_j < 0\}$; put $L := \beta_+ - \beta_- \geq 1$. From $\Delta_t = 0$ for $t \geq c$ we get:

(F) the forward recursion

$$G_{t+\beta_+} = -\rho^{-\alpha_{C_+}} \sum_{C \neq C_+} (-1)^{r-|C|} \rho^{\alpha_C} G_{t+\beta_C},$$

whose right-hand side involves only $G_{t+\beta_-}, \dots, G_{t+\beta_+-1}$;

(B) likewise $G_{t+\beta_-}$ is determined by $G_{t+\beta_-+1}, \dots, G_{t+\beta_+}$.

Let the state be $\mathbf{x}_t := (G_{t+\beta_-}, \dots, G_{t+\beta_+-1}) \in \mathcal{W}^L$ and let $\Sigma := \{\mathbf{x}_t : t \geq c\}$, a finite set. By (F), $\mathbf{x}_{t+1}$ is determined by $\mathbf{x}_t$, so $\Psi(\mathbf{x}_t) := \mathbf{x}_{t+1}$ is a well-defined map $\Sigma \rightarrow \Sigma$; by (B) it is injective on Σ . An injective self-map of a finite set is a bijection, so $(\mathbf{x}_t)_{t \geq c}$ is purely periodic: there is $q \geq 1$ with $\mathbf{x}_{t+q} = \mathbf{x}_t$ for all $t \geq c$. Reading off the first component, $G_{t+q} = G_t$ for $t \geq c + \beta_-$, that is $G(z+qu) = G(z)$ whenever $t(z) \geq c + \beta_-$. If $\Delta \equiv 0$ the same holds for every t , and G has the two independent periods qu and kv . $\square$

Proposition 8.9 (The first half-plane). *Let $\vartheta : \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ with $\vartheta = \sum_{i=1}^m \vartheta_i$, $m \geq 2$, where ϑ_i has period $h_i = k_i v_i$ with v_i primitive and the v_i pairwise non-parallel. Let R be a lattice-convex region on which ϑ is fully periodic, with vectors a, b , and put $K := K_R$. Then*

- (a) $a, b \in K$ and $\dim K = 2$;
- (b) for each i exactly one of the following holds:
 - (α) the line $\mathbb{R}v_i$ meets $\text{int } K$, and then ϑ_i is doubly periodic;
 - (β), (γ) otherwise there are $\sigma_i \in \{+, -\}$ and $t_i \in \mathbb{Z}$ such that ϑ_i is fully periodic on the half-plane $U_i := \{\sigma_i \pi_i \geq t_i\}$, and U_i contains a translate $R + w_i$ of R ;
- (c) if ϑ is non-periodic and no component is doubly periodic, then K is the sector of opening $< \pi$ spanned by two rays $\sigma_a v_a, \sigma_b v_b$ with $a \neq b$, and its interior contains no $\pm v_j$. In particular K cannot be a half-plane.

Proof. Step 1 (global extension). By Lemma 8.2, $a, b \in K$, $\dim K = 2$, $g := a + b \in \text{int } K$, and there is a doubly periodic $\tilde{\vartheta}$ with $\tilde{\vartheta} = \vartheta$ on R and $\text{Per}(\tilde{\vartheta})$ of finite index. Choose $N_0 \geq 1$ with $N_0 \mathbb{Z}^2 \subseteq \text{Per}(\tilde{\vartheta})$ and put $H_j := N_0 h_j$. Pick $x_0 \in C_R$, so that $x_0 + K \subseteq C_R$.

Step 2 (differentiating). Fix i and put $Q_i := \prod_{j \neq i} (T^{H_j} - 1)$, with formal exponent set $\mathcal{E}_i = \{H_C : C \subseteq [m] \setminus \{i\}\}$ and the expansion (0.2). If $z + \mathcal{E}_i \subseteq R$, then

$$(Q_i \vartheta)(z) = \sum_{C \subseteq [m] \setminus \{i\}} (-1)^{m-1-|C|} \tilde{\vartheta}(z + H_C) = \tilde{\vartheta}(z) \cdot (1-1)^{m-1} = 0,$$

because every H_C lies in $\text{Per}(\tilde{\vartheta})$. On the other hand H_j is a period of ϑ_j , so $Q_i \vartheta_j = 0$ for $j \neq i$, and therefore

$$D_i := Q_i \vartheta_i = Q_i \vartheta \quad \text{vanishes on } R'_i := \{z : z + \mathcal{E}_i \subseteq R\}.$$

Put $C' := \bigcap_{e \in \mathcal{E}_i} (C_R - e)$. Then $R'_i = C' \cap \mathbb{Z}^2$, and $C' \supseteq y_0 + K$ where $y_0 := x_0 + N'g$ with N' large: indeed for $x \in y_0 + K$ and $e \in \mathcal{E}_i$ one has $x - x_0 + e = N'g + e + (x - y_0) \in K$, because $N'g + e \in K$.

Step 3 (saturation along v_i). D_i has period h_i , so D_i vanishes on $\text{Sat}_i := \{z : (z + \mathbb{Z}h_i) \cap R'_i \neq \emptyset\}$. There are three cases.

(α) $\mathbb{R}v_i \cap \text{int } K \neq \emptyset$. Then v_i or $-v_i$ lies in $\text{int } K$, so for any z the point $z + nh_i$ enters $y_0 + K$ as $n \rightarrow +\infty$ or as $n \rightarrow -\infty$; hence $\text{Sat}_i = \mathbb{Z}^2$ and $D_i \equiv 0$. Apply Lemma 8.8 in its whole-plane case to the basis (v_i, u_i) , the hypothesis $t(H_j) = N_0 \det(v_i, h_j) \neq 0$ holding because the directions are pairwise non-parallel: ϑ_i is doubly periodic.

(β), (γ) $\mathbb{R}v_i \cap \text{int } K = \emptyset$. A convex set and a line missing its interior can be separated, so K lies in one of the two closed half-planes bounded by $\mathbb{R}v_i$; choose σ_i with $K \subseteq \{\sigma_i \pi_i \geq 0\}$. Since $\dim K = 2$, the functional $\sigma_i \pi_i$ does not vanish identically on K . For $\tau > 0$ the chord $K \cap \{\sigma_i \pi_i = \tau\}$ is either a segment in the direction v_i (case (γ): $K \cap \mathbb{R}v_i = \{0\}$, and the length is $\lambda\tau$ with $\lambda > 0$) or a ray (case (β): v_i is parallel to one of the boundary rays of K). In case (β) take any $c_i > 0$; in case (γ) take $c_i > 0$ with $\lambda c_i \geq |h_i|$. If $\sigma_i \pi_i(z - y_0) \geq c_i$, then

$$(z + \mathbb{R}v_i) \cap (y_0 + K) = y_0 + (K \cap \{\sigma_i \pi_i = \sigma_i \pi_i(z - y_0)\})$$

contains a segment of length $\geq |h_i|$ and hence a point of $z + \mathbb{Z}h_i$, these points being spaced $|h_i|$ apart along that line; so $z \in \text{Sat}_i$. Therefore D_i vanishes on the half-plane $\{\sigma_i \pi_i \geq t\}$ with $t := c_i + \sigma_i \pi_i(y_0)$. Apply Lemma 8.8 to the basis $(v_i, \sigma_i u_i)$, for which $t(z) = \sigma_i \pi_i(z)$ and $t(H_j) = \sigma_i N_0 \det(v_i, h_j) \neq 0$: the field ϑ_i is fully periodic on $U_i := \{\sigma_i \pi_i \geq t_i\}$ with $t_i := t + \beta_-$. Since $y_0 + K \subseteq C'$ and $K \subseteq \{\sigma_i \pi_i \geq 0\}$, the half-plane U_i contains a translate of R .

Step 4 (part (c)). Suppose ϑ is non-periodic and no component is doubly periodic. Then case (α) occurs for no i , that is, no line $\mathbb{R}v_i$ meets $\text{int } K$. Now K is a closed convex cone with $\dim K = 2$. If K were a half-plane, at most one line $\mathbb{R}v_i$ could lie on its boundary and all the others would fall into case (α), a contradiction; and $K = \mathbb{R}^2$ is excluded in the same way. So K is a sector of opening $< \pi$, each of its two boundary rays lies on one of the lines $\mathbb{R}v_a, \mathbb{R}v_b$ with $a \neq b$ — here $m \geq 2$ and the two boundary rays are not parallel — and its interior contains no $\pm v_j$. $\square$

Corollary 8.10 (Theorem 8.1(i)). *Let ϑ and R be as in Theorem 8.7, let $p > \max \mathcal{A}$ be a prime, and let $\vartheta = \sum_i \bar{\vartheta}_i$ be the decomposition of Lemma 8.5, with the h_i pairwise non-parallel. Apply Proposition 8.9 to each component and absorb the components falling into case (α), which are doubly periodic, into any other component by Operation A of Lemma 8.14; on the half-plane U_k of that component the sum is again the restriction of a doubly periodic field, hence still fully periodic. The resulting decomposition $\vartheta = \sum_{i \in I} G_i$ satisfies: $|I| \geq 2$, since ϑ is non-periodic; no component is doubly periodic; the periods are pairwise non-parallel; and each G_i is fully periodic on a half-plane $U_i = \{\sigma_i \pi_i \geq t_i\}$ whose boundary is parallel to h_i , with $U_i \supseteq R + w_i$. Moreover K_R is, by Proposition 8.9(c), the sector spanned by two adjacent rays $\sigma_a v_a, \sigma_b v_b$.*

Remark 8.11 (Relation to the body of the paper and to [4]). (1) Proposition 8.9 is the one-sided version of Lemma 4.1 of §4 — the support of $f_i = Q_i \eta$ lies in a strip — and Lemma 8.8 is the general form of step (6) of Lemma D.7. (2) Proposition 8.9(b) is the conclusion of [4, Proposition 3.1(i)], but obtained without induction, without ONED and without $\mathbb{Z}_p$ -minimality; consequently Lemma 2.10, Lemma 2.11 and Claim 3.2 of [4], and the two gaps (b) and (d) recorded in Remark 8.13, no longer enter the chain. (3) Proposition 8.9(c) is the content of [4, Claim 3.2]: in the oriented-edge convention of that paper, the recession cone of an (ℓ_i, ℓ_{i+m-1}) -region should be exactly the sector between the adjacent rays ℓ_{i+m-1} and $\ell_{i+m} = -\ell_i$; this may serve as a consistency check against [4]. (4) In (b) neither the non-periodicity of ϑ nor the minimality of the decomposition is required; components falling into case (α) are simply absorbed.

8.4. The second half-plane: the remaining external input.

Assumption 8.12 ([4, Proposition 3.1(ii)], weak form; that is, Theorem 8.1(ii)). Let ϑ be as in Theorem 8.7, let $\vartheta = \sum_i \vartheta_i$ be a $\mathbb{Z}$ -minimal periodic decomposition of it, of order $m = \text{ord}(\xi)$ as in Remark 8.3 and with pairwise non-parallel periods, and let p be the prime supplied for this

decomposition by [2, Theorem 1.6], so that the reduction $\vartheta = \sum_i \bar{\vartheta}_i$ modulo p is a $\mathbb{Z}_p$ -minimal decomposition of the same order. Apply Corollary 8.10 to it — Proposition 8.9 applies to any decomposition — and let $\vartheta = \sum_{i \in I} G_i$ and the half-planes U_i be the result. Then for every $i \in I$ there is a half-plane V_i with $V_i \cap U_i = \emptyset$ such that $G_i|_{V_i}$ is fully periodic.

By step (γ) of Lemma 8.14, every half-plane of full periodicity of an h_i -periodic field that is not doubly periodic is bounded by a line parallel to h_i ; so V_i necessarily lies in the complementary side $\{\sigma_i \pi_i < t_i\}$ of U_i , and shrinking either U_i or V_i destroys nothing. Assumption 8.12 is therefore insensitive to the exact position of the half-planes. In its Proposition 3.1(ii), [4] proves this assertion for a ϑ of the same provenance ([1, Lemma 4.6]) and for a $\mathbb{Z}_p$ -minimal decomposition of it; the half-plane its part (i) supplies lies on the same side of the region R as the one produced by Proposition 8.9, and one may shrink whichever of the two is the larger. We do not treat Assumption 8.12 as an established fact; it is written into the hypothesis of Theorem 8.15.

In the remark and the three paragraphs that follow we report what we checked in the sources. They use the notation of the cited papers (NEL, ONED, regions, the supports $\mathcal{S}_\varphi$, and the internal numbering of [4], [1] and [2]), which is not set up in this paper; nothing in §§0–7 or in the statement of Theorem 8.15 depends on them.

Remark 8.13 (Reading the proof in [4]). The chain of the proof of [4, Theorem 3.5] is: η non-periodic with a non-trivial annihilator $\Rightarrow$ (by [1, Theorem 1.9]) there is $\ell \in \text{NEL}(\eta)$ with $\pm \ell \in \text{ONED}(\eta) \Rightarrow$ (repeating the two cases of [1, Lemma 4.6], which use Propositions 2.13–2.14, Lemma 3.5, Lemma 4.2 and Claims 4.7–4.15 of that paper) one obtains a non-periodic $\vartheta \in X_\eta$ and an (ℓ, ℓ') -region on which ϑ is fully periodic $\Rightarrow$ (by [2, Theorem 1.5]) one chooses p for which the reduction modulo p of a $\mathbb{Z}$ -minimal decomposition is a $\mathbb{Z}_p$ -minimal decomposition of the same order $\Rightarrow$ (by Proposition 3.1) one obtains U_i and V_i . Theorems 8.6 and 8.7 of this paper correspond to the first two steps, Lemma 8.5 and Corollary 8.10 to the part of the third and fourth steps that produces U_i , and Assumption 8.12 to the part that produces V_i . The material new to [4] here is Lemma 2.10, Lemma 2.11 and Proposition 3.1. We went through these three line by line.

- (a) The layer propagation of Lemma 2.10 is correct: after the factor $X^{h_k} - 1$ has been removed, $\mathcal{S}_\varphi$ has no edge parallel to ℓ_k , so a single vertex can be placed outside the region and the propagation proceeds half-line by half-line.
- (b) Lemma 2.11 is correct, but its proof writes the translation along $v_{\ell''}$ as a multiple of the period u'' ; one should first pass to a subsequence lying in a fixed residue class modulo u'' and argue there.
- (c) Claim 3.2, which excludes $\iota' \neq \iota + m - 1$ using Lemma 2.11 and Proposition 2.6, is correct.
- (d) The inductive step of Proposition 3.1(i) applies the induction hypothesis to the partial sum $\vartheta_1 + \dots + \vartheta_{m-1}$ without checking that this partial sum satisfies the hypothesis $\ell_1, \ell_{m-1} \in \text{ONED}$. This is repairable: the proof of (i) uses the ONED hypothesis only in order to place the two asymptotic directions of the region on edges of $\mathcal{S}_\varphi$ (Lemma 2.9 and Claim 3.2), and the directions of the region of the partial sum are ℓ_1 and ℓ_{m-1} anyway, so it suffices to restate the induction hypothesis as “for every configuration of minimal order m' which is fully periodic on some $(\ell_\iota, \ell_{\iota+m'-1})$ -region”. The conclusion is unaffected. In this paper part (i) is replaced by Proposition 8.9, so that (a)–(d) no longer enter the chain.
- (e) Claim 3.3(a) deduces from Theorem 2.8 (the $\mathbb{Z}_p$ and convex form of Szabados’ theorem) and from Boyle–Lind that θ is ℓ_j -periodic; the passage “ θ_j fully periodic $\Rightarrow \vartheta_j$ fully periodic” is correct, because translating the h_j -periodic ϑ_j along ℓ_j produces only finitely

many translates. In Claim 3.3(b), the “maximal convex region L_α ”, the largest interval in the union of entire layers along ℓ_k containing a given strip, is well defined; there the translation $t_{n_\alpha} + s_\alpha \sigma t_k \tau$ has to be a positive multiple of τv_{ℓ_j} for the hypothesis of the Claim to apply, which does hold at the three places where Step 3 uses it ($j = p - 1$, $k = p$, $\tau = 1$ and $j = 2$, $k = 1$, $\tau = -1$), since $\sigma\tau = 1$ in each; but the general statement of the Claim does not record this constraint.

- (f) The angular argument of Step 3, that $v_{\ell_{p-1}}$ points into the interior of U_i for $i \leq p - 2$ and of V_i for $i \geq p + 1$, rests on $\ell_1, \dots, \ell_m$ lying in a half-circle and on $\ell_{i+m} = -\ell_i$, and is correct; the sector structure of Proposition 8.9(c) can be used directly to establish that ordering along the half-circle.

Assumption 8.12 depends on (e) and (f). Item (e) is a gap in the exposition and can be filled as described. We found no substantive error.

The check against [1] (the published version). [1, Theorem 1.9] reads: a configuration with $\mathcal{A} \subset \mathbb{Z}$ and a non-trivial annihilator is fully periodic provided that for every line ℓ either $-\ell \notin \text{ONED}$ or $\ell \notin \text{ONED}$. It carries *no* hypothesis that the non-periodic configurations in the orbit closure have the same order; that hypothesis appears only in [1, Theorem 1.10], a standing assumption of its §4.1, and in its Claims 4.11 and 4.16. When the proof of [4, Theorem 3.5] repeats the two cases of [1, Lemma 4.6] it uses only Claims 4.7–4.10 (first case) and 4.12–4.15 (second case), and the role of Claims 4.11 and 4.16 — locating the second asymptotic direction of the region — is taken over by its own Claim 3.2. Those claims, and Propositions 2.13–2.14, Lemmas 2.6–2.8, 3.5, 4.2 and 4.5, do not depend on the equal-order assumption. The proof of [1, Theorem 1.9] itself uses the periodic-limit theorem of Kari and Moutot [9] and the Boyle–Lind theorem. Theorem 8.7 above quotes [1, Lemma 4.6] directly, under the equal-order assumption supplied by Remark 8.3, and does not need the position of the second asymptotic direction; so either way of quoting the source is available.

The check against [2] (arXiv:2204.06658v1). The result cited as “[2, Theorem 1.5]” in [4] is Theorem 1.6 of that version: for $\mathcal{A} \subset \mathbb{Z}_+$ and a $\mathbb{Z}$ -minimal decomposition $\eta = \eta_1 + \dots + \eta_m$, there is a prime p with $\mathcal{A} \subset \llbracket p \rrbracket$ for which $\bar{\eta} = \bar{\eta}_1 + \dots + \bar{\eta}_m$ is a $\mathbb{Z}_p$ -minimal decomposition; this agrees with the way [4] restates it. Its proof chooses p so that the values of the finite-alphabet configurations $\varphi_i \eta_i + N_i \mathbb{N}$ lie in $\llbracket p \rrbracket$, shows first, by Proposition 2.9 of that paper with $R = \mathbb{Z}$, that no $\bar{\eta}_i$ is fully periodic, and then derives a contradiction from Lemma 3.1 there (if a shorter $\mathbb{Z}_p$ -decomposition existed, some $\bar{\eta}_i$ would be fully periodic). The proof of that Lemma 3.1 uses the fact that the reflected convex supports of two annihilators have no parallel edges to conclude $\text{ONED} = \emptyset$, then Boyle–Lind for full periodicity, peeling off one component at a time; the proof of Proposition 2.9 for an arbitrary R (its Lemma 2.8, the “division by $X^v - 1$ ” construction, together with its Proposition 2.7) we also read through. Neither contains an error. The other results of [2] quoted in [4] — its Lemma 2.4, its Proposition 2.7, a rewriting of [5, Lemma 5.10], and its Proposition 2.9 — match in number and in content. In this paper that theorem is used only for the choice of the prime p in Assumption 8.12, that is, to align the decomposition used here with the $\mathbb{Z}_p$ -minimal decomposition of [4]; the proof of Theorem 8.1(i) does not need it.

The one link that is given only as a sketch. Claim 3.3(a) of [4], and through it its Proposition 3.1(ii) and the existence of the half-planes V_i , uses its Theorem 2.8, the $\mathbb{Z}_p$ and convex form of Szabados’ theorem that a minimal decomposition of order two does not have low convex complexity. We checked where that version is to be found: it is *not* in [2]; the published version of [1] states, as its Theorem 1.7, only the $\mathbb{Z}$ -coefficient convex form, citing Szabados [15], whose own statement is over $\mathbb{Z}$ and for rectangles; the arXiv version of [1] carries a remark that the same proof works over a finite field; and [4] gives a sketch of the rewriting (a pigeonhole argument producing agreement with a translate on a strip, its Proposition 2.7 of [2] giving ONED

in both directions for two edge directions, and the propagation of balanced sets from Cyr–Kra [5] and [15, Lemma 4]). Every step of the sketch concerns only pattern counting for finite-alphabet configurations and the geometry of generating sets, and none of it involves the arithmetic of the coefficient ring, so that the $\mathbb{Z}_p$ form is plausible; but nowhere is it written out. Theorem A does not depend on Theorem 2.8; §8 depends on it indirectly, through Assumption 8.12. **Appendix D gives a complete proof of [4, Theorem 2.8]** (Theorem D.1 and Corollary D.2, using nothing beyond the one-dimensional Morse–Hedlund theorem), so that this link is no longer a sketch and Claim 3.3(a) of [4] may quote Theorem D.1 directly.

8.5. Normalisation.

Lemma 8.14 (Normalisation). *Let $\theta: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ be non-periodic and suppose $\theta = \sum_{i=1}^m G_i$, where each $G_i: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ has a non-zero period h_i and there are disjoint half-planes U_i, V_i on which G_i is fully periodic. Then θ is a star configuration.*

Call a family $(G_i, h_i, U_i, V_i)_{i=1}^m$ with these properties an *admissible decomposition* of θ , of length m . Since θ is non-periodic, every admissible decomposition has $m \geq 2$: for $m = 1$ we would have $\theta = G_1$ periodic.

Proof. Step (γ): alignment of the components that are not doubly periodic. Let G have a non-zero period h , not be doubly periodic, and be fully periodic on disjoint half-planes U with normal n and V with normal n' . *Claim:* $\langle h, n \rangle = 0$. Let $\tilde{G}$ be the global extension of $G|_U$ from Lemma 8.2 and put $\Gamma_U := \text{Per}(\tilde{G})$, of finite index. Suppose $\langle h, n \rangle \neq 0$; replacing h by $-h$ we may assume $\langle h, n \rangle > 0$. For $z \in U$ we have $z + h \in U$, so $\tilde{G}(z + h) = G(z + h) = G(z) = \tilde{G}(z)$: the Γ_U -periodic function $\tilde{G}(\cdot + h) - \tilde{G}$ vanishes on U . But a Γ_U -periodic function vanishing on a half-plane vanishes everywhere: Γ_U contains some $N'\mathbb{Z}^2$ and hence a vector g with $\langle g, n \rangle > 0$, and any z is translated into U by $z + Ng$ for N large. So $\tilde{G}$ has period h . Now for any z pick N with $z + Nh \in U$: then $G(z) = G(z + Nh) = \tilde{G}(z + Nh) = \tilde{G}(z)$, that is $G = \tilde{G}$ is doubly periodic, a contradiction.

Hence $n \perp h$. Write $h = kv$ with v primitive and $k \geq 1$, after changing the sign of h if necessary, and put $\pi = \det(v, \cdot)$. Since $v^\perp = (-v_2, v_1)$ satisfies $\langle z, v^\perp \rangle = \pi(z)$ and $n \parallel v^\perp$, we have $\langle z, n \rangle = \kappa \pi(z)$ for some $\kappa \in \mathbb{R}^\times$, so U is either $\{\pi < \ell\}$ or $\{\pi > r\}$ with $\ell, r \in \mathbb{Z}$. The same applies to V . If U and V were both of the type $\{\pi < \cdot\}$, or both of the type $\{\pi > \cdot\}$, they would intersect; so one is $\{\pi < \ell\}$ and the other is $\{\pi > r\}$, and disjointness says exactly $\ell \leq r + 1$. Taking $\tilde{L}, \tilde{R}$ to be the global extensions of G on these two half-planes, G satisfies (S1) with period kv , (S2), and (S3).

Operation A: absorbing a doubly periodic component. Suppose G_i is doubly periodic and $m \geq 2$; pick $k \neq i$. Since $\text{Per}(G_i)$ has finite index, $\text{Per}(G_i) \cap \mathbb{Z}h_k$ contains some $h'_k \neq 0$, and $G'_k := G_k + G_i$ has period h'_k . On U_k we have $G'_k = \tilde{G}_k + G_i$ by Lemma 8.2, the restriction of a doubly periodic field, hence fully periodic; likewise on V_k . Deleting G_i and replacing G_k by G'_k gives an admissible decomposition of length $m - 1$.

Operation C: merging parallel components. Suppose G_i, G_k with $i \neq k$ are both not doubly periodic and $h_i \parallel h_k$. By Step (γ) there is a primitive v with $h_i = \pm k_i v$, $h_k = \pm k_k v$, and $\{U_i, V_i\} = \{\{\pi < \ell_i\}, \{\pi > r_i\}\}$, $\{U_k, V_k\} = \{\{\pi < \ell_k\}, \{\pi > r_k\}\}$, where $\pi = \det(v, \cdot)$. Then $G_i + G_k$ has period $\text{lcm}(k_i, k_k)v$, equals $\tilde{L}_i + \tilde{L}_k$ on $\{\pi < \min(\ell_i, \ell_k)\}$ and $\tilde{R}_i + \tilde{R}_k$ on $\{\pi > \max(r_i, r_k)\}$, both restrictions of doubly periodic fields and hence fully periodic; and these two half-planes are disjoint, since $\min(\ell_i, \ell_k) \leq \ell_i \leq r_i + 1 \leq \max(r_i, r_k) + 1$. Merging gives an admissible decomposition of length $m - 1$.

Termination and conclusion. Starting from the given admissible decomposition, repeat: if some component is doubly periodic, apply Operation A; otherwise, if some pair is parallel, apply Operation C. Each step strictly decreases the length, so the process terminates. At termination no component is doubly periodic, the periods are pairwise non-parallel, and $m \geq 2$. By Step (γ) every component satisfies (S1)–(S3) and the primitive directions v_i are pairwise non-parallel. So θ is a star configuration. $\square$

8.6. The convex Nivat conjecture.

Theorem 8.15 (Convex Nivat conjecture, conditional on Assumption 8.12). *Assume Assumption 8.12, that is the weak form of [4, Proposition 3.1(ii)], which is Theorem 8.1(ii). Let $\xi: \mathbb{Z}^2 \rightarrow \mathcal{A}$ with $\mathcal{A}$ finite, and suppose there is a non-empty finite lattice-convex S with $P_\xi(S) \leq |S|$. Then ξ is periodic.*

Proof. Suppose the conjecture has a counterexample. Embed the alphabet into $\mathbb{Z}_{>0}$ by an injective encoding, which changes neither the number of patterns nor periodicity, and use Remark 8.3 to choose a counterexample of minimal ord, still written ξ ; it is non-periodic and has low convex complexity.

(0) *The annihilator.* Theorem 8.4 supplies pairwise non-parallel $h_1, \dots, h_m$ and $\varphi = \prod_i (X^{h_i} - 1)$ annihilating ξ ; we may take $m = \text{ord}(\xi)$.

(1) *The region.* Theorem 8.6 supplies ℓ , and Theorem 8.7 a non-periodic $\vartheta \in X_\xi$ together with a lattice-convex region R on which ϑ is fully periodic.

(2) *The decomposition.* Take a $\mathbb{Z}$ -minimal periodic decomposition $\vartheta = \sum_i \vartheta_i$, of order $m = \text{ord}(\xi)$ by Remark 8.3 and with pairwise non-parallel periods, and take p as Assumption 8.12 requires, by [2, Theorem 1.6]. Then Lemma 8.5(b) gives $\vartheta = \sum_i \bar{\vartheta}_i$ over $\mathbb{F}_p$ with each $\bar{\vartheta}_i$ of period h_i .

(3) *The first half-plane.* Corollary 8.10 gives $\vartheta = \sum_{i \in I} G_i$ with $|I| \geq 2$, no component doubly periodic, pairwise non-parallel periods, and each G_i fully periodic on U_i .

(4) *The second half-plane.* Assumption 8.12 gives half-planes V_i disjoint from U_i with $G_i|_{V_i}$ fully periodic. At this point the whole of Theorem 8.1 holds.

(5) *The complexity does not increase.* Since $\mathcal{A} \subset \llbracket p \rrbracket$, reading the values of ϑ in $\mathbb{F}_p$ merges no colours: as an $\mathbb{F}_p$ -valued configuration, ϑ has the same pattern set and the same periodicity as it has as an $\mathcal{A}$ -valued one. Every finite-window pattern of a configuration in X_ξ is a pattern of ξ , a limit in the product topology agreeing with some translate on any finite window. Hence

$$P_{\vartheta}(S) \leq P_\xi(S) \leq |S|. \quad (8.1)$$

(6) *Normalisation.* ϑ is non-periodic, $\vartheta = \sum_{i \in I} G_i$ over $\mathbb{F}_p$ with each G_i of period h_i and fully periodic on the disjoint half-planes U_i, V_i . By Lemma 8.14, ϑ is a star configuration.

(7) *Contradiction.* By Theorem 7.3, $P_{\vartheta}(S) \geq |S| + 1$, contradicting (8.1). $\square$

Corollary 8.16 (Nivat's conjecture, under the hypothesis of Theorem 8.15). *Under the same hypothesis: if $P_\xi([1, n] \times [1, k]) \leq nk$ for some $n, k \geq 1$, then ξ is periodic. (A rectangle is lattice-convex.)*

Remark 8.17 (The link that is not yet removed). Assumption 8.12 is the only dependence left between this paper and [4]. A route that avoids [4] reaches, at present, the following point. Let $c \in I$ and $\sigma \in \{+, -\}$, and put

$$J(c, \sigma) := \{j \neq c : \sigma_j \pi_j(\sigma h_c) < 0\}.$$

Translating along $\sigma N h_c$, where $N\mathbb{Z}^2 \subseteq \bigcap_j \text{Per}(\tilde{G}_j)$, and passing to a subsequential limit $\Xi = \lim T^{n_k \sigma N h_c} \vartheta \in X_\xi$, one gets

$$\Xi = G_c + \sum_{j \notin J} \tilde{G}_j + \sum_{j \in J} G_j^\infty,$$

where $\tilde{G}_j$ is the global extension of $G_j|_{U_j}$. When $|J(c, \sigma)| = 1$, the configuration Ξ is a sum of two periodic configurations with different directions, so Theorem D.1 makes Ξ periodic, and over $\mathbb{F}_p$ one deduces further that G_j^∞ is doubly periodic, j being the unique element of J : along such directions, every subsequential limit of G_j on the side where it is not yet known is doubly periodic. What is still missing is a propagation lemma from “every limit at infinity is doubly periodic” to “the far field is itself eventually doubly periodic”; the one-dimensional example $\dots (0^3 1)^{N_3} (0^2 1)^{N_2} (01)^{N_1}$ shows that the information carried by the limits alone does not suffice. Claim 3.3 of [4] fills that step with the propagation of balanced sets of Cyr and Kra. We leave it to future work.

APPENDIX A. SKELETON OF THE PROOF

- §1 $\exists a : DI_a \neq 0 \implies$ Theorem 7.3 at once, no convexity needed; otherwise $DI_a \equiv 0$ and $D\eta = 0$.
- §2 $\Lambda_i = \text{exceptional spectrum}, A = \prod_i A_i(T^{v_i}), Z = \text{Newt}(A);$
 $f(T)\eta = \text{const} \implies A \mid f \implies \dim U_S \geq |S| - |R_Z(S)| + 1.$
- §3 Ae agrees on adjacent realised sectors $\implies Ae_\theta = b$, doubly periodic, all H_i -invariant.
- §4 $f_i = Q_i \eta \neq 0$ and transversally bounded $\implies \exists d : J(d, \cdot) = D_z[\eta(z)\eta(z+d)] \neq 0.$
- §5 $A(T_d)J = A(T_z T_d^{-1})J = 0; R/(a, c)$ spanned by $Y^{Z-Z} \implies d \in (Z - Z) \cap \mathbb{Z}^2.$
- §6 $(Z - Z) \cap \mathbb{Z}^2 = (Z \cap \mathbb{Z}^2) - (Z \cap \mathbb{Z}^2) \implies d = q_1 - q_0, q_0, q_1 \in Z.$
- §7 $|R_Z(S)|$ quadratic characters independent modulo $U_S \implies P_\theta(S) \geq |S| + 1.$
- §8 $[8], [1] \xrightarrow{\text{Prop. 8.9}} U_i \xrightarrow{\text{Asm. 8.12, [4]}} V_i \xrightarrow{\text{Lem. 8.14}} \text{star configuration}$
 $\xrightarrow{\text{Thm. 7.3}} \text{convex Nivat} \implies \text{Nivat}.$

Dependencies: §§0–7 are self-contained, using only the standard facts that $\mathbb{C}[T^\pm]$ is a unique factorisation domain, Newton-polygon additivity (Ostrowski), the Chinese remainder theorem, the Nullstellensatz, and the existence of unimodular triangulations of lattice polygons. The external inputs to §8 are Kari and Szabados [8] and Colle [1], both published, together with the one link that has not been refereed: Assumption 8.12, that is [4, Proposition 3.1(ii)], the second half-plane. The first half-plane is proved unconditionally here, in Proposition 8.9; the result [4, Theorem 2.8] on which [4] rests in proving that proposition is proved in full in Appendix D; and [2, Theorem 1.6] is used only to align the decomposition of this paper with the $\mathbb{Z}_p$ -minimal decomposition of [4].

APPENDIX B. INDEPENDENT EDGE-INCREMENT RESULTS, NOT PART OF THE PROOF CHAIN

Before the argument of §§0–8 took its present form, the same question was approached along a different route, by comparing $P_\theta(S)$ with $P_\theta(S \setminus E)$ for an edge E of the polygon $\text{Conv}(S)$ and controlling the increment through the combinatorics of the edge directions of $\text{Conv}(S)$ against the directions v_i . That route produced a number of statements of independent interest: an exact identity for the lattice width of $\text{Conv}(S)$ in a given direction in terms of its edge data, a divisibility relation between the determinants of three consecutive edge directions, a propagation

principle for sums of functions each of which depends on a single one of the projections π_i and which vanish on enough consecutive layers, a description of the periods surviving after repeated differencing, a count of the far-field patterns along an edge, and lower bounds of the form $P_\theta(S) - P_\theta(S \setminus E) \geq 2$ for short edges and of the form $P_\theta(S) - P_\theta(S \setminus E) \geq$ (a row-length quantity) in general. None of them is used above, their proofs are not reproduced here, and the notation they require is not set up in this paper; they will be treated separately. Nothing in §§0–8 depends on any of them.

APPENDIX C. NUMERICAL VERIFICATION

The scripts are `star.py` (construction, enumeration and all checks), `verify.py` and `verify2.py` (the two randomised experiments), and `lemma31.py` (the counterexample of §C.3). They depend only on `numpy` and are supplied as ancillary files together with their output for the stated seeds; every number below is read off that output.

C.1. Method.

- *Configurations.* Each component is specified by a primitive direction v_i , a tangential period k_i , an arbitrary value table on the transition strip $[\ell_i, r_i]$, and left and right tails L_i, R_i given, in the basis (v_i, u_i) of §0.3, as doubly periodic tables in $(t \bmod a, s \bmod k_i)$. Axioms (S1) and (S3) hold by construction. Axiom (S2) is decided exactly: F_i is doubly periodic if and only if it coincides with the global extension of its left tail (a doubly periodic function vanishing on a half-plane vanishes identically), and that coincidence is tested on the strip and on one full period beyond it. With M the least common multiple of all the k_j and of all the tail period indices, the lattice Γ of §0.3 contains $M\mathbb{Z}^2$, and the program uses $H_i = Mk_i v_i$; by Remark 1.4 this choice, which need not be the minimal one, does not affect the Case A/Case B decision, and it does not affect Λ_i either, since refining the period only adds zero Fourier coefficients.
- *Pattern counts.* The S -patterns are enumerated at all anchors of the square box of radius $R = 90$ and again of radius $R = 180$. Away from the crossing region of the strips the patterns along each strip are H_i -periodic and, inside a sector, $M\mathbb{Z}^2$ -periodic, so the count is exact as soon as the box is larger than the crossing region plus one period plus $\text{diam}(S)$; the two counts agreed on every window of every configuration reported below.
- *Deciding the case.* DI_a is computed on the box of radius 40 from the formal expansion (0.1); all-zero for every colour a means Case B.
- *Spectrum and zonotope.* The fields $\Xi_i^\pm$ and $G_i^{\sigma, \epsilon}$ are built as in §2.1; for each difference field $d_{i, \sigma, \epsilon, a}$ a length- κ_i DFT is taken row by row over all rows within $2M + 2$ of the strip (outside the strip the rows repeat with period M), and the union of the frequencies with coefficient of modulus above 10^{-9} gives Λ_i, d_i and Z . The set $R_Z(S)$ is decided by testing whether all subset sums of the $d_i v_i$, translated by r , lie in $\text{Conv}(S)$.
- *Ranks.* The encoding w is a random injection into $\{1, \dots, 100\}$. The rank of the $|\mathcal{L}| \times (|S| + 1)$ matrix $(1, w(\gamma(s)))_{\gamma \in \mathcal{L}, s \in S}$ is computed over $\mathbb{F}_q$ with $q = 2^{31} - 1$ and reported as $\dim U_S$; it is a lower bound for the rational rank, which is the direction needed to verify Lemma 2.5. In Case B with $R_Z(S) \neq \emptyset$ the matrix is extended by the columns $\Phi_r(\gamma) = w(\gamma(r + q)) w(\gamma(r + q + d))$, $r \in R_Z(S)$, of §7, and the rank $H_2(S)$ of the extended matrix is compared with $\dim U_S + |R_Z(S)|$ (Lemma 7.2) and with $|S| + 1$ (Theorem 7.3).
- *Witness and background.* In Case B, for every non-zero $d \in (Z - Z) \cap \mathbb{Z}^2$ the function $J(d, \cdot)$ is computed on the box of radius 40 and the non-zero instances are collected (Proposition 5.3); the first of them, with a q as in Corollary 6.2, is the pair (d, q) used

for the Φ_r . The polynomial A is built with complex coefficients from the Λ_i , and Ae_θ is checked to be $M\mathbb{Z}^2$ -periodic on the box (Proposition 3.3).

C.2. Results. Experiment 1 (script `verify.py`, seed 1). We generated 84 random star configurations with $p \in \{2, 3, 5\}$, $m \in \{2, 3\}$, directions drawn without repetition from the 8 primitive vectors $(1, 0), (0, 1), (1, \pm 1), (1, \pm 2), (2, \pm 1)$, $k_i \in \{1, 2, 3\}$, tail periods of 1 or 2 rows, strip widths 1 to 3 with $\ell_i \in \{-1, 0, 1\}$, all tables uniformly random over $\mathbb{F}_p$ subject to (S2); component 1 has independently drawn left and right tails, the other components have $L_i = R_i$. Windows: the 2×3 , 3×3 , 2×4 and 4×2 rectangles, the 6-point triangle with vertices $(0, 0), (2, 0), (0, 2)$, the 8-point hexagon with vertices $(1, 0), (2, 0), (3, 1), (2, 2), (1, 2), (0, 1)$, the 5-point rhombus with vertices $(0, 0), (1, 1), (2, 0), (1, -1)$, and the 3-point segment $\{0, 1, 2\} \times \{0\}$. Result: 81 configurations in Case A and 3 in Case B; $P_\theta(S) \geq |S| + 1$ held on all 8 windows of all 84 configurations. The three Case-B instances have $d_i \in \{3, 6\}$, so Z is larger than every window and $R_Z(S) = \emptyset$ throughout; on them $\dim U_S$ was exactly $|S| + 1$ on every window, the witness d existed (192, 310 and 28 non-zero d respectively), and Ae_θ was doubly periodic to within $2 \cdot 10^{-15}$. These are instances of Remark 2.6; Experiment 2 targets $R_Z(S) \neq \emptyset$.

Experiment 2 (script `verify2.py`, seed 7). We generated 157 configurations of the form $\sum_i w_i \circ \pi_i$, so $k_i = 1$ and constant tails: each $w_i: \mathbb{Z} \rightarrow \mathbb{F}_p$ takes a constant value c_i outside a window of length 1 to 4 and arbitrary values inside it, with $L_i = R_i = c_i$; $p \in \{2, 3, 5\}$, $m = 2$ with probability 0.7 and $m = 3$ with probability 0.3, directions drawn from the 11 primitive vectors $(1, 0), (0, 1), (1, \pm 1), (1, \pm 2), (2, \pm 1), (1, \pm 3), (3, 1)$. Windows: the 3×4 , 4×4 , 5×3 and 4×3 rectangles, the 15-point triangle with vertices $(0, 0), (4, 0), (0, 4)$, the 17-point hexagon with vertices $(1, 0), (3, 0), (4, 2), (3, 4), (1, 4), (0, 2)$, the 13-point rhombus with vertices $(0, 0), (2, 2), (4, 0), (2, -2)$, and the 12-point parallelogram with vertices $(0, 0), (3, 0), (5, 2), (2, 2)$. Result: 128 in Case A and 29 in Case B, all with $d_i = 1$; $P_\theta(S) \geq |S| + 1$ held on all 8 windows of all 157 configurations, and every check listed in §C.1 passed on every Case-B instance. Of the 29 Case-B configurations, 14 have $R_Z(S) \neq \emptyset$ for at least one window, giving 35 configuration-window pairs with $R_Z(S) \neq \emptyset$ and $|R_Z(S)| \in \{1, 2, 3, 4, 5\}$; all 35 have $m = 2$ (no Case-B instance with $m = 3$ and $R_Z(S) \neq \emptyset$ was produced by this generator). Representative rows of the output:

p	directions	S	$ S $	$P_\theta(S)$	d_i	$ R_Z(S) $	$\dim U_S$	$ S + 1 - R_Z(S) $	$H_2(S)$	d
3	$(1, 2), (1, 0)$	4×4	16	35	$(1, 1)$	4	13	13	17	$(-1, -1)$
3	$(1, 2), (1, 0)$	hexagon	17	37	$(1, 1)$	5	13	13	18	$(-1, -1)$
3	$(1, 2), (1, -2)$	hexagon	17	45	$(1, 1)$	3	15	15	18	$(-1, -1)$
5	$(1, -3), (1, 0)$	4×4	16	37	$(1, 1)$	2	15	15	17	$(-1, 1)$
2	$(2, -1), (1, 2)$	hexagon	17	41	$(1, 1)$	2	16	16	18	$(-2, 0)$
5	$(2, 1), (1, 2)$	4×4	16	54	$(1, 1)$	1	16	16	17	$(-2, -2)$

On all 35 pairs with $R_Z(S) \neq \emptyset$ the value of $\dim U_S$ was *exactly* $|S| + 1 - |R_Z(S)|$ (over $\mathbb{F}_q$), so the space of affine relations consists of exactly the multiples of A and Lemma 2.5 is tight; and $H_2(S)$ was exactly $\dim U_S + |R_Z(S)| = |S| + 1$, so the quadratic characters make up the deficit exactly, as in §7. On the 116 two-component configurations of Experiment 2 the criterion of Observation C.1 below agreed with the computed case in all 116 instances, and the Case-B direction pairs had $|\det(v_1, v_2)| \in \{2, 3, 4, 5, 6, 7, 10\}$.

Observation C.1 (A criterion for Case B when $m = 2$). Suppose $m = 2$ and both components satisfy $L_i = R_i$; write $\delta_i = F_i - L_i$ for the deviation field, supported in strip i . Replace H_i by a sufficiently large multiple, which by Remark 1.4 does not change the Case A/Case B decision, and expand the four-point template $\{z, z + H_1, z + H_2, z + H_1 + H_2\}$ at an anchor z : δ_1 is constant along the H_1 -pairs and δ_2 along the H_2 -pairs, so $De_\theta(z) = 0$ if and only if at least one of the two

pairs of constants agrees. Hence *Case B holds if and only if* $\text{supp } \delta_1 \cap \text{supp } \delta_2 = \emptyset$, that is, no point is deviated by both components at once. In particular, when $k_i = 1$, the tails are constant and equal, and $|\det(v_1, v_2)| = 1$, one is always in Case A: the map $(\pi_1, \pi_2): \mathbb{Z}^2 \rightarrow \mathbb{Z}^2$ is then surjective and the two deviation sets must meet. Case B occurs only for $|\det| \geq 2$. The sample of Experiment 2 was generated at random and then sorted by the numerical test; the criterion explains why Case B appeared only on direction pairs with $|\det| \geq 2$.

C.3. A counterexample to the generalisation of Lemma 3.1. (`lemma31.py`.) Take $p = 2$, $m = 3$, $v_1 = (1, 0)$, $v_2 = (0, -1)$, $v_3 = (1, -1)$. Tails: $L_1 = 0$ and $R_1 = \mathbb{K}[x \text{ even}]$; $L_2 = L_3 = 0$ and $R_2 = R_3 = \mathbb{K}[3 \mid x]$; the transition strip of each component is constant 1, of width 1. Here $\pi_2(v_1) = \pi_3(v_1) = 1 > 0$, so $B_1^+ = R_2 + R_3 \equiv 0$ and $B_1^- = L_2 + L_3 = 0$, and the program computes $\Lambda_1 = \{\pm 1\}$ (with $\kappa_1 = 12$ and frequencies $\{0, 6\}$), so that $A_1(T^{v_1}) = T^{2v_1} - 1$.

- For the pairs of realised sectors adjacent along the rays $\pm v_1$, namely (R_1, R_2, R_3) with (L_1, R_2, R_3) and (R_1, L_2, L_3) with (L_1, L_2, L_3) : the difference after applying $A_1(T^{v_1})$ is 0, up to a floating-point error of $1.3 \cdot 10^{-16}$, for both colours.
- For the mixed choice (R_1, R_2, L_3) against (L_1, R_2, L_3) — here $B = R_2 + L_3 = \mathbb{K}[3 \mid x]$, and the sign vector $(+, +, -)$ corresponds to the cone $\{y > 0, x > 0, x + y < 0\} = \emptyset$, which is not realised: the difference after applying $A_1(T^{v_1})$ has maximum absolute value 2.

By hand: along the x -direction, $\Delta_1 = \mathbb{K}[R_1 + B = 1] - \mathbb{K}[L_1 + B = 1]$ takes the values $(-1, 0, 1, 0, 1, 0)$ for $x \bmod 6 = 0, \dots, 5$, whereas its translate by $x \mapsto x + 2$ is $(1, 0, 1, 0, -1, 0)$. The two differ, so Δ_1 is not killed by $T^{2v_1} - 1$: its spectrum contains cube roots of unity, and those are not in Λ_1 . This is why §3 can only assert that Ae is constant across the *realised* sectors, where $B = B_i^e$ — and that is exactly, and only, what Lemma 3.2 needs.

APPENDIX D. THE $\mathbb{F}_p$ AND CONVEX FORM OF SZABADOS' THEOREM

This appendix proves [4, Theorem 2.8] in full, and so removes the one step which Remark 8.13 reports as being, in the whole chain, the only one given as a sketch in the source. The proof follows Szabados [15] — pigeonhole, balanced sets, and the one-dimensional Morse–Hedlund theorem — with two changes. The coefficient ring is $\mathbb{F}_p$: every argument below uses only equality of values and the fact that a certain difference is periodic in a given direction, and none of it touches the arithmetic of the ring. And the window is an arbitrary lattice-convex set instead of a rectangle: in Lemma D.5 one deletes, at each step, whichever of the two extreme rows is the shorter. In addition, Lemma 3 of [15] rests on Lemma 39 of Kari and Szabados [8]; we replace it by two self-contained arguments, Lemma D.3, which uses the explicit shape of the annihilator, and step (6) of Lemma D.7, which uses the two-sided recursion supplied by the generating property. Apart from the one-dimensional Morse–Hedlund theorem, nothing external is quoted here.

D.0. The statements.

Theorem D.1. *Let p be a prime, let $\theta_1, \theta_2: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ have periods $h_1, h_2 \in \mathbb{Z}^2 \setminus \{0\}$ respectively, with h_1 and h_2 linearly independent over $\mathbb{R}$, and put $\theta = \theta_1 + \theta_2$. If $P_\theta(S) \leq |S|$ for some non-empty finite lattice-convex S , then θ is periodic.*

Corollary D.2 ([4, Theorem 2.8]). *Let $\eta: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ and let $\eta = \eta_1 + \eta_2$ be an $\mathbb{F}_p$ -minimal periodic decomposition, of order 2 — that is, a decomposition into periodic $\mathbb{F}_p$ -configurations with as few summands as possible, here two. Then η does not have low convex complexity.*

Proof. Order 2 forces η to be non-periodic, a periodic configuration having a minimal decomposition of length 1, and it forces the two periods to point in different directions, components with parallel periods being mergeable into one, against minimality. If η had low convex complexity, Theorem D.1 would make it periodic, a contradiction. $\square$

Claim 3.3(a) of [4] applies its Theorem 2.8 to $\theta = (\sum_{i \neq k} \theta_i) + \theta_k$, the two summands having periods $\tilde{h}_j \in \ell_j$ and h_k in different directions; Theorem D.1 applies to that verbatim.

D.1. Notation. Let $\mathbf{u}$ be the primitive vector in the direction of h_1 , so that $h_1 = c_1 \mathbf{u}$ with $c_1 \geq 1$, and let $\pi := \det(\mathbf{u}, \cdot) : \mathbb{Z}^2 \rightarrow \mathbb{Z}$, which is surjective. The *rows* are the fibres $L_t := \pi^{-1}(t)$; neighbouring lattice points of a row differ by $\mathbf{u}$. Put $\Delta := \pi(h_2) \neq 0$ and, replacing h_2 by $-h_2$ if necessary, assume $\Delta > 0$. For integers $a \leq b$ the *strip* $\text{St}[a, b] := \{v : a \leq \pi(v) \leq b\}$, of *width* $b - a$, is the union of the rows $L_a, \dots, L_b$ and is invariant under $\pm \mathbf{u}$. For a non-empty finite $F \subset \mathbb{Z}^2$ put $\text{ht}(F) := \max \pi(F) - \min \pi(F)$; F *fits into* $\text{St}[a, b]$ if some translate of it is contained there, that is if $\text{ht}(F) \leq b - a$. For a non-empty finite lattice-convex F write $E_+(F) := F \cap L_{\max \pi(F)}$ and $E_-(F) := F \cap L_{\min \pi(F)}$, the top and bottom rows; lattice convexity makes every non-empty row of F a run of consecutive lattice points.

For a finite $F \subset \mathbb{Z}^2$ write $\text{Pat}(F, \theta) := \{\theta|_{v+F} : v \in \mathbb{Z}^2\}$, the patterns being regarded as functions $w \mapsto \theta(v + w)$ on F , so that $P_\theta(F) = |\text{Pat}(F, \theta)|$ in the notation of §0.2, with $P_\theta(\emptyset) = 1$; X_θ is the orbit closure, so that every finite-window pattern of an $x \in X_\theta$ is a pattern of θ . “ $x = \theta$ on U ” means $x(v) = \theta(v)$ for all $v \in U$. Inside this appendix the letter D denotes auxiliary finite sets and never the difference operator of §0.3, which does not occur here.

D.2. Two algebraic facts. For $s \in \mathbb{Z}$, the h_2 -periodicity of θ_2 gives

$$\theta(v + sh_2) - \theta(v) = \theta_1(v + sh_2) - \theta_1(v), \quad \text{which is } h_1\text{-periodic in } v. \quad (\text{D.1})$$

Consequently: if $T^{sh_2}\theta$, that is $v \mapsto \theta(v + sh_2)$, agrees with θ on a set F , then it agrees with θ on $F + \mathbb{Z}h_1$.

Lemma D.3 (Periodicity on a strip implies periodicity). *Suppose $b - a \geq \Delta$ and $q \geq 1$, and that $\theta(v + qu) = \theta(v)$ for all $v \in \text{St}[a, b]$. Then θ has period qh_1 .*

Proof. The strip is invariant under $+\mathbf{u}$, so $\theta(v + qh_1) = \theta(v)$ for $v \in \text{St}[a, b]$, applying the hypothesis c_1 times. Put $\psi(v) := \theta(v + qh_1) - \theta(v)$; since θ_1 has period h_1 we have $\psi(v) = \theta_2(v + qh_1) - \theta_2(v)$, so ψ has period h_2 , and $\psi = 0$ on $\text{St}[a, b]$. Write $h_2 = gh'_2$ with h'_2 primitive and take any lattice line $L = \{v_0 + sh'_2 : s \in \mathbb{Z}\}$ in the direction h'_2 . Then $\psi|_L$ is periodic in s with period g , while $\pi(v_0 + sh'_2) = \pi(v_0) + s\pi(h'_2)$ runs through an arithmetic progression of common difference $\pi(h'_2) = \Delta/g \geq 1$; a closed interval $[a, b]$ of length $b - a \geq \Delta = g \cdot (\Delta/g)$ contains at least g of its terms, so L meets $\text{St}[a, b]$ in at least g consecutive of its points. A function of period g vanishing at g consecutive points vanishes identically, so $\psi|_L \equiv 0$. As L was arbitrary, $\psi \equiv 0$. $\square$

D.3. Balanced sets.

Definition D.4. Let B be a non-empty finite lattice-convex set, let $\sigma \in \{+, -\}$ and put $E := E_\sigma(B)$. Call B σ -balanced if

- (i) $P_\theta(B) \leq |B|$;
- (ii) $P_\theta(B) < P_\theta(B \setminus E) + |E|$;
- (iii) every non-empty row of B contains at least $|E| - 1$ points.

Lemma D.5 (Existence). *Let S be a non-empty finite lattice-convex set with $P_\theta(S) \leq |S|$. Then there are a sign σ and a σ -balanced set $B \subseteq S$.*

Proof. Construct $D_0 := S \supsetneq D_1 \supsetneq \cdots \supsetneq D_K = \emptyset$ as follows: if $D_i \neq \emptyset$, choose σ_i with $|E_{\sigma_i}(D_i)| \leq |E_{-\sigma_i}(D_i)|$ — either sign will do when D_i has a single row, where $E_+ = E_- = D_i$ — and put $D_{i+1} := D_i \setminus E_{\sigma_i}(D_i)$. Each D_i is again lattice-convex: after an extreme row has been deleted the remaining points lie in an open half-plane, so the lattice points of their convex hull still belong to D_i and do not lie on the deleted row. Put $f(i) := P_\theta(D_i) - |D_i|$; then $f(0) \leq 0$ and $f(K) = 1$, so there is a smallest i with $f(i+1) > 0$, and for it $f(i) \leq 0 < f(i+1)$. Let $B := D_i$, $\sigma := \sigma_i$ and $E := E_\sigma(B)$. Condition (i) is $f(i) \leq 0$. For (ii), $P_\theta(B) \leq |B| = |B \setminus E| + |E| < P_\theta(B \setminus E) + |E|$.

For (iii), write $n = |E|$ and $E' = E_{-\sigma}(B)$, so $|E'| \geq n$. For a real $t \in [\min \pi(B), \max \pi(B)]$ let $\lambda(t)$ be the length of the intersection of the convex polygon $\text{Conv}(B)$ with the line $\{\pi = t\}$; λ is concave, by the concavity of the length of parallel chords of a convex set. On the two extreme rows that intersection is exactly $\text{Conv}(E)$, respectively $\text{Conv}(E')$, a supporting line meeting the polygon in the convex hull of the vertices it carries, of lengths $(n-1)|u|$ and $(|E'| - 1)|u| \geq (n-1)|u|$. Hence $\lambda(t) \geq (n-1)|u|$ for every t . A closed segment of length at least $(n-1)|u|$ on a lattice line whose consecutive points are $|u|$ apart contains at least $n-1$ lattice points, and these belong to B by lattice convexity. $\square$

D.4. Ambiguity strips and the propagation lemma.

Definition D.6. Say that θ has the $+$ -ambiguity strip $\text{St}[a, b]$, where $a \leq b$, if there is an $x \in X_\theta$ with $x = \theta$ on $\text{St}[a, b-1]$ — an empty condition when $a = b$ — and $x \neq \theta$ at some point of L_b . The $-$ -ambiguity strips are defined symmetrically, replacing u by $-u$.

Lemma D.7 (Propagation). *Let B be $+$ -balanced and put $n := |E_+(B)|$. If θ has some $+$ -ambiguity strip $\text{St}[a, b]$ into which B fits, that is with $b - a \geq \text{ht}(B)$, then θ is periodic.*

Proof. (1) *Placing B .* Translate B so that $\max \pi(B) = b$; then $B \subseteq \text{St}[a, b]$, and $E := E_+(B) = B \cap L_b = \{e_1, \dots, e_n\}$ with $e_{l+1} = e_l + u$, while $B_0 := B \setminus E \subseteq \text{St}[a, b-1]$, possibly empty. Put $t_- := \min \pi(B)$. Take a witness $x \in X_\theta$ of the ambiguity: $x = \theta$ on $\text{St}[a, b-1]$ and $x \neq \theta$ somewhere on L_b . Note that $B_0 + iu \subseteq \text{St}[a, b-1]$ for every $i \in \mathbb{Z}$, so $x = \theta$ on each $B_0 + iu$.

(2) *Two determination rules.* For $0 \leq i \leq n$ put $D_i := B_0 \cup \{e_1, \dots, e_i\}$ and $D'_i := B_0 \cup \{e_{n-i+1}, \dots, e_n\}$. By (ii) of Definition D.4, $P_\theta(D_n) - |D_n| < P_\theta(D_0) - |D_0|$, so there is a $k \in \{0, \dots, n-1\}$ with $P_\theta(D_{k+1}) - |D_{k+1}| < P_\theta(D_k) - |D_k|$, that is $P_\theta(D_{k+1}) < P_\theta(D_k) + 1$; the restriction map $\text{Pat}(D_{k+1}, \theta) \rightarrow \text{Pat}(D_k, \theta)$ being onto, this forces $P_\theta(D_{k+1}) = P_\theta(D_k)$ and makes that map a bijection. Hence:

(R₊) for every $y \in X_\theta$ and $v \in \mathbb{Z}^2$, the value $y(e_{k+1} + v)$ is determined by $y|_{D_k+v}$.

In the same way there is a $k' \in \{0, \dots, n-1\}$ with

(R₋) for every $y \in X_\theta$ and $v \in \mathbb{Z}^2$, the value $y(e_{n-k'} + v)$ is determined by $y|_{D'_{k'}+v}$.

(3) *Claim:* $x|_{D_k+iu} \neq \theta|_{D_k+iu}$ for every $i \in \mathbb{Z}$. Suppose $x = \theta$ on $D_k + i_0u$. If $k = 0$, then $D_0 = B_0$, and (1) together with (R₊) gives $x(e_1 + iu) = \theta(e_1 + iu)$ for every i , that is $x = \theta$ on L_b , against the choice of the witness. So let $k \geq 1$. By (R₊), $x(e_{k+1} + i_0u) = \theta(e_{k+1} + i_0u)$; hence $x = \theta$ on $D_k + (i_0 + 1)u$, whose points are those of $B_0 + (i_0 + 1)u$, by (1), together with the points $e_l + (i_0 + 1)u = e_{l+1} + i_0u$ for $1 \leq l \leq k$, which lie in $D_k + i_0u$ when $l \leq k-1$ and are the point just treated when $l = k$. By induction $x = \theta$ on $D_k + iu$ for every $i \geq i_0$, hence on the half-line $\{e_1 + su : s \geq i_0\} \subset L_b$. Now go left. If $k' = 0$, then (1) and (R₋) give $x = \theta$ on $\{e_n + iu : i \in \mathbb{Z}\} = L_b$, a contradiction. If $k' \geq 1$, choose i_1 with $n - k' + i_1 \geq i_0$; the points $e_1 + (n - k' + i_1)u, \dots, e_1 + (n - 1 + i_1)u$ of the top row of $D'_{k'} + i_1u$ then lie on that half-line, so $x = \theta$ on $D'_{k'} + iu$ for every $i \geq i_1$. By (R₋), $x(e_{n-k'} + i_1u) = \theta(e_{n-k'} + i_1u)$, whence $x = \theta$ on $D'_{k'} + (i_1 - 1)u$, whose top-row points $e_l + (i_1 - 1)u = e_{l-1} + i_1u$, $n - k' + 1 \leq l \leq n$, either

lie in $D'_{k'} + i_1\mathbf{u}$ or are the point just treated. Descending induction gives $x = \theta$ on $D'_{k'} + i\mathbf{u}$ for every $i \in \mathbb{Z}$, and the top rows of these sets cover all of L_b , against the choice of the witness. The claim follows.

(4) *Counting.* By (3) and (1), for each i the patterns $x|_{B+i\mathbf{u}}$ and $\theta|_{B+i\mathbf{u}}$ are two B -patterns of θ which agree on $B_0 + i\mathbf{u}$ and differ somewhere. So every pattern γ in $\Gamma := \{\theta|_{B_0+i\mathbf{u}} : i \in \mathbb{Z}\} \subseteq \text{Pat}(B_0, \theta)$ has at least two extensions to a B -pattern, $N(\gamma) \geq 2$. But

$$\sum_{\gamma \in \text{Pat}(B_0, \theta)} (N(\gamma) - 1) = P_\theta(B) - P_\theta(B_0) < |E| = n$$

by (ii), so $|\Gamma| \leq n - 1$; in particular $n \geq 2$, since otherwise Γ would be empty, which it is not.

(5) *Periodicity of the rows of B_0 .* Assume $B_0 \neq \emptyset$; when $B_0 = \emptyset$ this step is vacuous and one takes $Q_0 := 1$. For each row L_t of B_0 , $t_- \leq t \leq b - 1$, condition (iii) makes $B \cap L_t$ a run of $m_t \geq n - 1$ consecutive points $w_t, w_t + \mathbf{u}, \dots, w_t + (m_t - 1)\mathbf{u}$. Put $\xi_t(s) := \theta(w_t + s\mathbf{u})$ for $s \in \mathbb{Z}$. As i varies, $\theta|_{B_0+i\mathbf{u}}$ takes at most $n - 1$ values, so its restriction to the first $n - 1$ points of row t — that is, the word of length $n - 1$ read in ξ_t from position i — takes at most $n - 1$ values, and ξ_t has at most $n - 1$ subwords of length $n - 1$. By the one-dimensional Morse–Hedlund theorem [11] (a bi-infinite sequence with at most m subwords of some length m is periodic; see also [1, Theorem 1.1]), ξ_t is periodic. Let Q_0 be a common multiple of the periods of these rows; then $\theta(v + Q_0\mathbf{u}) = \theta(v)$ for all $v \in \text{St}[t_-, b - 1]$.

(6) *Extending upwards one row at a time.* Suppose $R \geq b$ and θ has period $Q\mathbf{u}$ on $\text{St}[t_-, R - 1]$, which for $R = b$ is step (5) with $Q = Q_0$. Choose w with $\pi(w) = R - b$, so that the top row of $B + w$ lies on L_R and $B_0 + w + i\mathbf{u} \subseteq \text{St}[t_-, R - 1]$; then the pattern $\theta|_{B_0+w+i\mathbf{u}}$ depends only on $i \bmod Q$. Put $y_s := \theta(e_1 + w + s\mathbf{u})$ and apply (R_+) and (R_-) to $y := \theta$ and $v = w + i\mathbf{u}$:

$$y_{i+k} \text{ is determined by } (i \bmod Q; y_i, \dots, y_{i+k-1}),$$

$$y_{i-1} \text{ is determined by } (i \bmod Q; y_i, \dots, y_{i+k'-1}),$$

the second being (R_-) at $v = w + (i - n + k')\mathbf{u}$. Put $K := \max(k, k', 1)$ and let $\sigma_i := (i \bmod Q; y_i, \dots, y_{i+K-1}) \in \mathbb{Z}/Q \times \mathbb{F}_p^K$. Then σ_{i+1} is determined by σ_i , and so is σ_{i-1} . Let Φ be the forward map, defined on the set Σ of states that actually occur; backward determinacy makes Φ injective on Σ , since $\Phi(\sigma) = \Phi(\sigma')$ forces $\sigma = \sigma'$, and Σ is finite, so Φ is a bijection of Σ and the bi-infinite orbit (σ_i) is periodic. Its period Q' is a multiple of Q , by the phase component, and $y_{s+Q'} = y_s$. Hence θ has period $Q'\mathbf{u}$ on $\text{St}[t_-, R]$. Iterating from $R = b$ up to $R = t_- + \Delta$, finitely many steps, gives a Q^* for which θ has period $Q^*\mathbf{u}$ on the strip $\text{St}[t_-, t_- + \Delta]$, of width Δ .

(7) By Lemma D.3, θ has period Q^*h_1 . □

Remark D.8. Step (5) of Lemma D.7 is the argument of [15, Lemma 4] and of [5, Lemma 2.24]; steps (6) and (7) replace [15, Lemma 3], which appeals to Lemma 39 of [8], and use nothing beyond the generating property and the explicit annihilator.

D.5. Proof of Theorem D.1. Suppose θ is not periodic. Replacing h_2 by ch_2 for an integer $c \geq 1$, which changes neither π nor $\mathbf{u}$, we may assume $\Delta \geq \text{ht}(S) + 1$. Put

$$D := \{\alpha h_1 + \beta h_2 : \alpha, \beta \in [0, 1)\} \cap \mathbb{Z}^2, \quad D_j := D + jh_2, \quad S_j := \text{St}[j\Delta, (j+1)\Delta - 1].$$

Every $v \in \mathbb{Z}^2$ is uniquely $v = \alpha h_1 + \beta h_2$ with $\alpha, \beta \in \mathbb{R}$, and $\pi(v) = \beta\Delta$, so $v \in S_j$ if and only if $\beta \in [j, j+1)$; in that case $v - \lfloor \alpha \rfloor h_1 = \{\alpha\}h_1 + \beta h_2$ is a lattice point of D_j . Hence $S_j = D_j + \mathbb{Z}h_1$, and distinct values of $\lfloor \alpha \rfloor$ give distinct points, so the union is disjoint. Put $N := P_\theta(D) < \infty$.

(a) *A balanced set, and unambiguity.* Take a σ -balanced $B \subseteq S$ by Lemma D.5. We may assume $\sigma = +$: otherwise replace u by $-u$, π by $-\pi$ and h_2 by $-h_2$, which keeps Δ positive, all the definitions being symmetric. Since θ is not periodic, the contrapositive of Lemma D.7 says: *for all $a \leq b$ with $b - a \geq \text{ht}(B)$ and every $x \in X_\theta$, if $x = \theta$ on $\text{St}[a, b - 1]$ then $x = \theta$ on $\text{St}[a, b]$.* Iterating row by row,

$$x = \theta \text{ on } \text{St}[a, b], \quad b - a \geq \text{ht}(B) - 1 \implies x = \theta \text{ on } \{\pi \geq a\}. \quad (\text{D.2})$$

(b) *A witness of non-periodicity.* $N!h_2$ is not a period of θ , so there is a v_0 with $\theta(v_0 + N!h_2) \neq \theta(v_0)$. Put $k := \lfloor \pi(v_0)/\Delta \rfloor - N$, so that $\pi(v_0) \geq (k + N)\Delta$.

(c) *Pigeonhole.* Among the $N + 1$ patterns $\theta|_{D_{k+j}}$, $0 \leq j \leq N$, at most N are distinct, so there are $0 \leq j < j' \leq N$ with $\theta(v + (j' - j)h_2) = \theta(v)$ for all $v \in D_{k+j}$. By (D.1) this holds for all $v \in S_{k+j}$, that is $x := T^{(j'-j)h_2}\theta \in X_\theta$ satisfies $x = \theta$ on $S_{k+j} = \text{St}[(k + j)\Delta, (k + j + 1)\Delta - 1]$. That strip has width $\Delta - 1 \geq \text{ht}(S) \geq \text{ht}(B)$, so (D.2) gives $x = \theta$ on $\{\pi \geq (k + j)\Delta\}$, that is

$$\theta(v + (j' - j)h_2) = \theta(v) \quad \text{whenever } \pi(v) \geq (k + j)\Delta.$$

Since $\pi(h_2) = \Delta > 0$, that half-plane is invariant under $+(j' - j)h_2$; applying the identity repeatedly, and using $(j' - j) \mid N!$, we get $\theta(v + N!h_2) = \theta(v)$ for every v with $\pi(v) \geq (k + j)\Delta$. But $\pi(v_0) \geq (k + N)\Delta \geq (k + j)\Delta$, so $\theta(v_0 + N!h_2) = \theta(v_0)$, contradicting (b). $\square$

Remark D.9. The proof of Theorem D.1 works over any finite alphabet for a sum of two periodic configurations, all that is used being (D.1), that the difference $\theta(\cdot + sh_2) - \theta$ is h_1 -periodic. Over $\mathbb{Z}$ it is the convex generalisation of [15, Theorem 1] recorded as [1, Theorem 1.7].

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